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Transformative impact of edge computing with 6G network using cloud computing

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Abstract

The upcoming 6G network and the quick development of edge computing promise to completely change the cloud computing market by meeting the increasing needs for real-time analytics, ultra-low latency, and huge data processing. Edge computing allows for the distribution of processing resources closer to the data source, hence reducing bottlenecks in centralized cloud infrastructures. For the upcoming generation of applications, such as autonomous systems, immersive AR/VR, and IoT-driven smart environments, the 6G network will provide a smooth, fast environment with its expected high bandwidth, low latency, and ubiquitous connection. The disruptive synergy between edge computing and 6G technology is examined in this essay, along with its significant impact on cloud computing. We look at the possibilities and difficulties of increasing the effectiveness of data processing, boosting security, and opening up new business models in cloud services. This study explores how edge computing, when combined with 6G networks, can revolutionize the well-established field of cloud computing.

Keywords Cloud computing, Edge computing, 6G networks, Latency reduction, High speed connectivity, Internet of thing, Augmented reality (AR), Virtual Reality (VR)

Introduction

Wireless communication's prolonged existence, especially since 6G first appeared, and edge computing's rapid growth [1]. The cloud computing ecosystem will change as a result of edge computing. The existing fourth-generation (4G) infrastructure has been highly beneficial to businesses involved in online entertainment and communication. The Internet of Vehicles (IoV), the Internet of Everything (IoE), and applications for virtual and augmented reality (AR/VR) will be integrated into future 5G communication networks. But the absence of human interaction in 5G communications has made companies and researchers look ahead to the next wave of wireless communication. 5G technology is devoid of human interaction despite its many advantages [2, 3]. The 6G generation's connectivity foundation turns connected objects into related intelligence. Research and development expenditures for the 6G communication system are mostly focused on the product, the press, and the essential components [4].

As the 5G networks have moved into the commercial deployment phase, several international research organizations have focused on the 6G networks. With ultra-low latency (microseconds) and peak data rates of around 1 Tbps, 6G networks are designed to enhance performance, as illustrated in Fig. 1. Furthermore, the 6G network aims to increase capacity by approximately 1000 times that of the 5G network by utilizing terahertz frequencies and spatial multiplexing. Global coverage will be facilitated by efficiently integrating satellite and underwater

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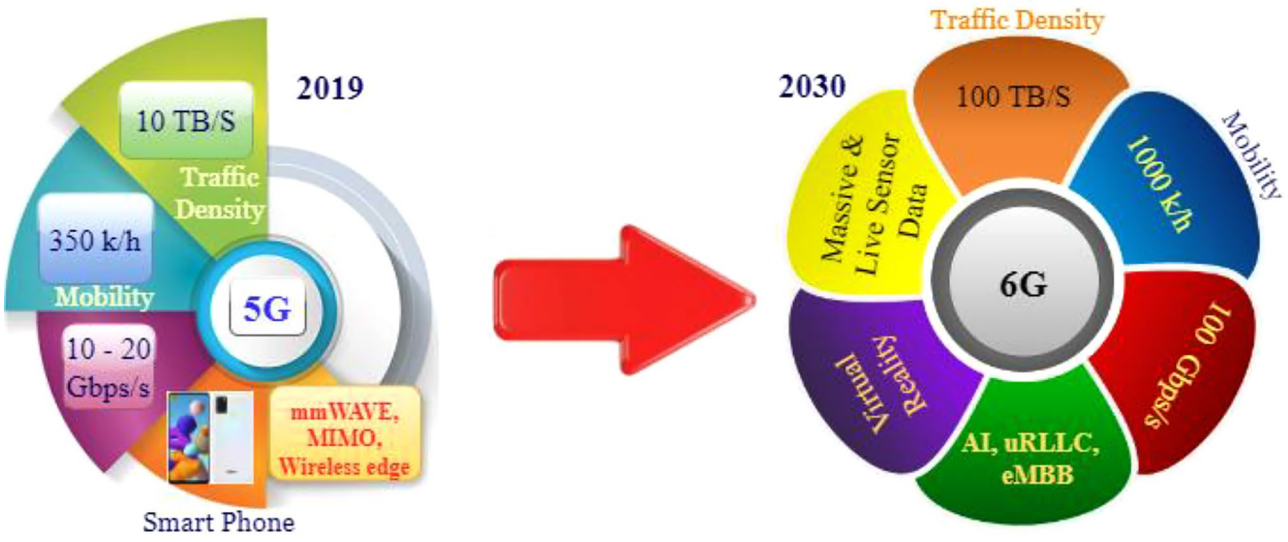


Fig. 1 The shifting communication technology 5 G to 6G

communication networks in 6G networks [5]. The ubiquitous mobile ultra-broadband (uMUB), ultrahigh-speed with low-latency communications (uHSLLC), and ultra-high data density (uHDD) are three further innovative classifications for 6G networks [6].

In [7], the concept of AI-enhanced 6G wireless networks is introduced. The 6G network architecture and its uses for various AI-powered smart service applications are also detailed. The article also delves into the communication challenges posed by 6G for large-scale distributed machine learning (ML) in mobile artificial intelligence (AI) applications. Figure 1 shows the 5 G communication technology shifting to 6G technology for fast, reliable, secure communication. The autonomy, self-configuration, and self-healing, of 6G wireless communication will be greatly improved by AI and ML. With 6G-enabled technology, reducing latency, improving real-time processing, and ultra-reliable low-latency connectivity will be simple to accomplish. The 6G communication technology four pillar will support all required challenges in terms of energy efficiency, connectivity, physical technology, and edge AI. The detailed 6G enabled communication components architecture is illustrated in Fig. 2. The 6G communication technology's four pillars will support all required challenges regarding energy efficiency, connectivity, physical technology, and edge AI. Tariq et *al.* provide extensive detail about using 6G, enabling technologies, and research challenges [8].

The 6G is service-centric, feasible ultra-sensitive applications, reliable, AI-enabled, and supports virtual and augmented reality and 100 Gbps data rates [3, 9]. The objective of this study is to present an expert opinion on the most innovative and popular research paths that may influence the development of 6G technology along with edge node integration impact on multi cloud environment.

Table 1 shows a comprehensive list of acronyms and their meanings used throughout the paper.

Motivation

The motivation for exploring the integration of edge computing with 6G networks in cloud/multi-computing stems from the need to overcome existing latency, bandwidth, and data processing efficiency limitations.

Traditional cloud computing is often delayed due to the distance between users and data centers. However, 6G's promise of ultra-low latency and high speeds, combined with edge computing's localized data processing, can significantly enhance real-time performance for applications such as autonomous vehicles and augmented reality. Additionally, the explosion of data from IoT devices and sensors necessitates efficient handling, which edge computing addresses by processing data closer to its source and easing the burden on central servers. This integration not only boosts reliability and reduces energy consumption but also supports the development of new applications, driving advancements in technology and contributing to economic and societal benefits.

Contribution

This study aims to draw a complete image of 6G technology on integrating edge computing and significantly cloud/multi computing through several vital contributions. It fosters architectural innovation by developing new distributed computing models and enhancing data storage and network management. Performance optimization is another critical area, as this synergy improves latency, throughput, and reliability for demanding

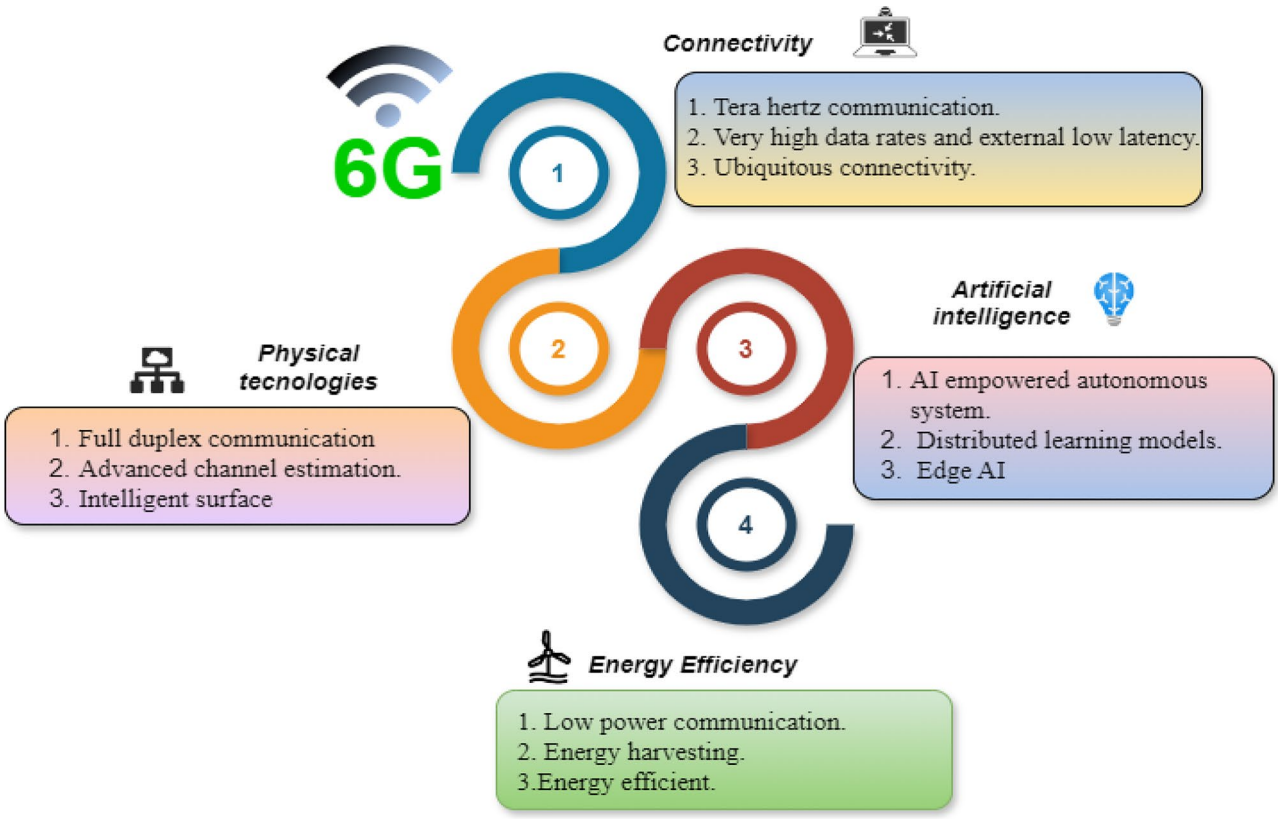


Fig. 2 The 6G enabled communication components

applications. Security and privacy protocols also evolve to address the new challenges posed by this integration, ensuring robust protection. Economically, it offers potential cost savings and new business models by reducing latency and increasing efficiency. Additionally, this integration drives the development of standardized frameworks for interoperability across various systems. It benefits society by supporting smart infrastructure, improving healthcare through remote diagnostics, and enabling interactive educational tools.

- Combining Edge Computing with 6G Networks: Investigation of the ways in which edge computing and 6G networks work in tandem to improve cloud computing services' efficiency, scalability, and performance.
- Better Real-Time Data Processing and Lower Latency: Analysis of how real-time data processing and quicker decision-making are made possible by 6G networks' reduced latency and edge computing's close proximity to data sources.
- Improved Privacy and Data Security: Studying the security and privacy effects of integrating 6G with edge computing, with a focus on decentralized edge processing and data storage.
- Assistance for Complex Use Cases in Cloud Computing: Identifying new applications made possible by the convergence of edge, 6G, and cloud, including industrial automation, smart cities, IoT ecosystems, and augmented and virtual reality (AR/VR).
- The study also analysed edge centric technological open challenges, implementation analysis and discussion.

Organization of paper

This article's remaining section is organized as follows: Related work is included in Section "State Of Art". Impact of 6G on cloud computing is presented in Section "Impact of 6G on cloud computing" and followed by 6G Enabled Use Cases & Key Enabled Technology for Edge/Cloud Computing in Section "6G enabled use cases & key enabled technology for Edge/Cloud computing".

Vision of Edge Computing with 6G Network on Cloud Computing is explored in Section "Vision Of Edge Computing With 6g Network On Cloud Computing", Section "Futurestics 6g applications", followed by Futuristic 6G Applications Section "Edge centric technological Open challenges, implementation analysis and discussion". Edge centric technological open challenges, implementation analysis & discussion, empirical results, and

Table 1 List of common acronyms and Description used in the paper

S. No.	Acronym	Meaning
1	6G	Sixth-Generation
2	IoV	Internet of Vehicles
3	IoE	Internet of Everything
4	AR/VR	Virtual and Augmented Reality
5	uMUB	ubiquitous mobile ultra-broadband
6	uHSLLC	Ultrahigh-speed with low-latency communications
7	uHDD	ultrahigh data density
8	ML	Machine Learning
9	AI	Artificial Intelligence
10	SOTA	State of the art
11	EC	Edge Computing
12	CC	Cloud Computing
13	URLLC	Ultra-reliable and low-latency communications
14	MBRLLC	Mobile broadband reliable low latency communications
15	ERLLC	Extremely reliable low-latency communications
16	RAN	Radio access network
17	DP	Differential Privacy
18	CDS	Cache Decision System
19	SB	Smart Buildings
20	BD	Big Data
21	HPC	High-performance computing
22	RL	Reinforcement learning
23	DDPG	Deep deterministic policy gradient
24	FL	Federated learning
25	IoT	Internet of Things
26	ITU	International Telecommunication Union
27	mMTC	massive machine type communication
28	eMB	enhanced mobile broadband
29	QoS	Quality-of-service
30	IIoT	Industry IoT
31	AIoT	Artificial Intelligence of Things
32	DT	Digital Twin
33	DTN	Digital twin network
34	NFT	Non fungible token
35	DL	Deep Learning
36	RAT	Radio Access Technologies
37	F-RANs	Fog radio access networks
38	CU	Centralized unit
39	UE	user equipment
40	XR	Extended Reality

future research direction, and conclusion are presented in Section "Emptical results", Section "Future research directions", and "Conclusion" respectively. The detailed structure of this paper is presented in Fig. 3. The structure of the study shows the distribution of the paper and the key areas covered in each section.

State of art

This section provides a state of the art (SOTA) of the dimensions and aspects of 6G and the transformative impact of edge computing (EC) with a 6G network on cloud computing (CC). The methodology adopted for SOTA is presented in Fig. 4. We started by reviewing the literature on significant databases, including Google, IEEE, Springer, Elsevier, and Delnet. The effect of 6G characteristics on edge, cloud and multi-computing is also explored.

Figure 5 shows the evolution of wireless communication with a timeline from 1G to 6G. Short-distance device-to-device communication, satellite, and terrestrial networking will all be included in the comprehensive, ever-widening 6G network. IoE devices also have less processing power and are smart and intelligent, able to make wise decisions. As an outcome, we need transfer data from the cloud to the edge and end device. To cut down on processing latency, we must move the process closer to the end devices in terms of cloud/fog. One of the biggest obstacles is coordinating data processing between edge devices and the cloud while controlling latency. From the healthcare sector to smart city applications, the convergence of edge computing with 6G networks heralds a paradigm shift in the processing, storing, and transmission of data. The following significant areas provide insight into this transformation:

6G enhanced ultra-reliable and low-latency services

One of the key communication services of the generation (6G) is ultra-reliable and low-latency communications (URLLC); nonetheless, for next-generation quality of services, reduced latency and improved real-time processing with delays as tiny as one millisecond, 6G offers ultra low latency, facilitating real-time processing and decision-making. Improved ultra-reliable and low-latency communications (eURLLC) 6G will encompass full connection with enhanced low latency for the IoE. Figure 6 shows the connectivity for edge node, IoE, and IoV for 6G eURLLC [11]. Mobile broadband reliable low latency communications (MBRLLC) is a generalization of enhanced mobile broadband (eMBB) and traditional ultra-reliable, low latency communications (URLLC).

Energy efficiency is critical to MBRLLC due to its impact on rate and reliability. Additionally, 6G devices have limited resources to support applications such as wireless Brain-Computer Interface (BCI), connected robotics and autonomous systems (CRAS), and extended reality (XR), among others [12]. Low latency, massive connectivity, and high reliability, are few of the key QoS characteristics that the massiveURLLC (mURLLC) in 6G simultaneously meets. Extremely reliable low-latency communications (ERLLC) are designed to provide low

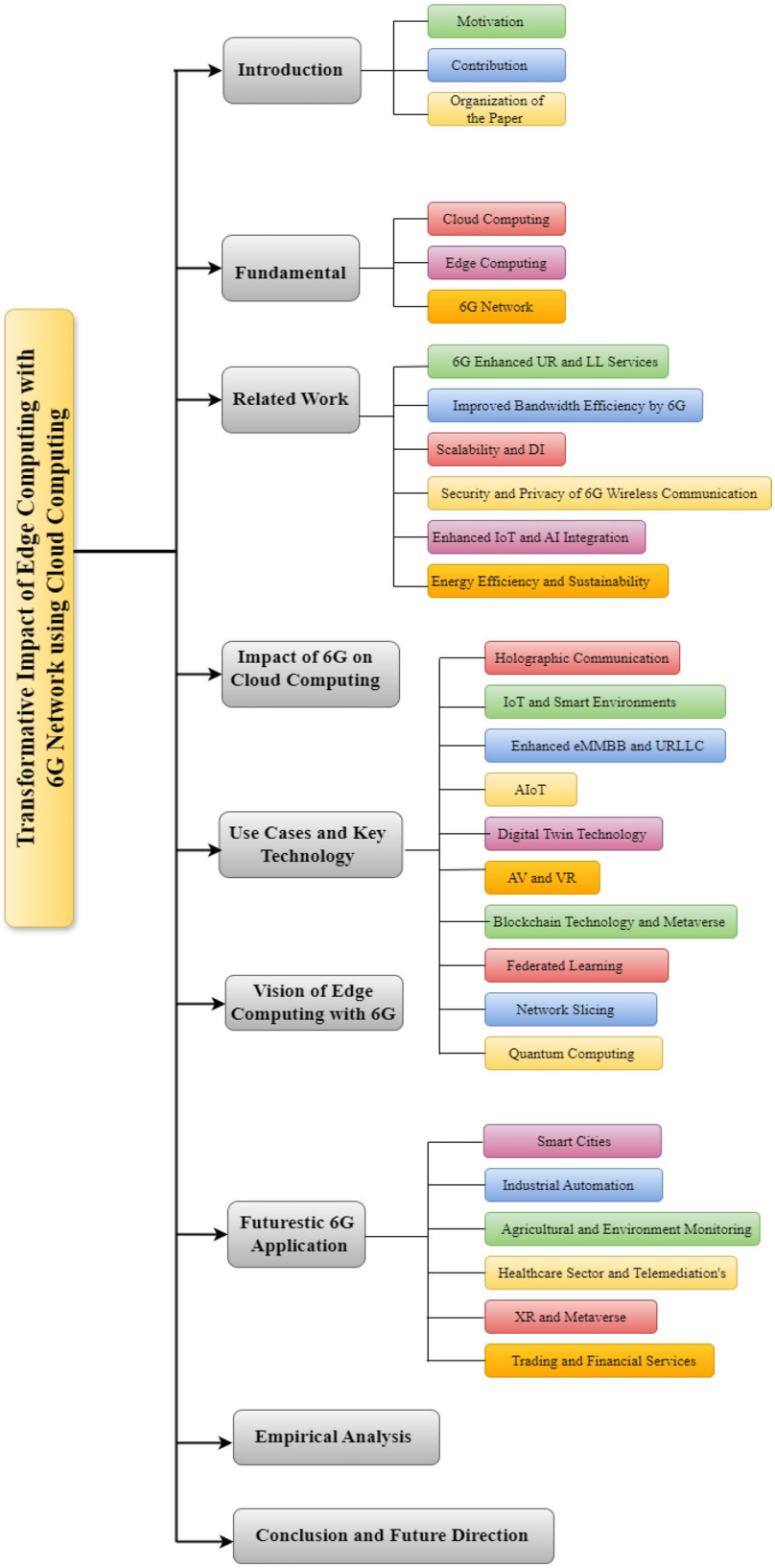


Fig. 3 Section-wise structure of paper

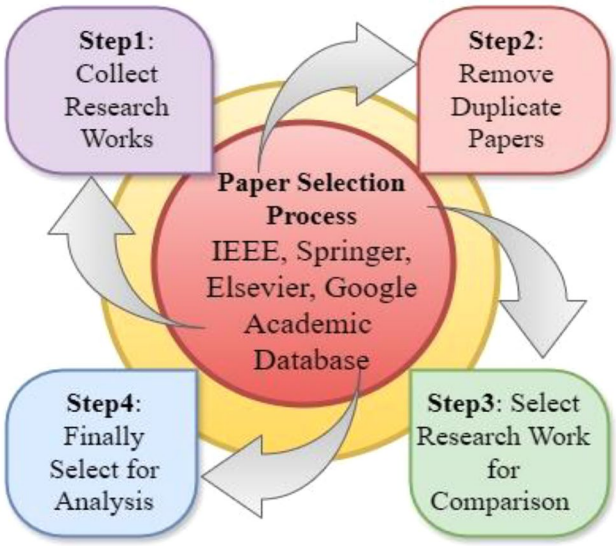


Fig. 4 Methodology applied for SOTA

latency along with reliability for IoE, IoV, edge devices, etc.

In traditional CC, data is transferred to centralized data centres, which frequently results in issue with latency, especially for real-time applications like remote surgery or autonomous vehicles. With 6G EC, data processing is moved closer to the source, greatly lowering latency and opening up previously implausible real-time applications.

Improved bandwidth efficiency by 6G

The intranet usage data in terms of bandwidth consumption and data rate presented in Fig. 7 depicts the significant rise in global internet usage (GB) from 10% in 2020 to 47% in 2028. With greater bandwidth capacities, 6G technology can handle data transmission speeds up to 100 times faster than 5 G by utilizing the distributed radio access network (RAN) and the terahertz (THz) spectrum. Furthermore, it can enhance spectrum sharing, reduce latency, and enhance capacity [10].

Local data processing and filtering made possible by EC ensures that only processed or necessary data is delivered to the cloud. By doing this, a large amount of traffic from the core network is offloaded, increasing bandwidth and enhancing the general effectiveness of cloud services. As a result, there is less stress on cloud infrastructure and better utilization of network resources.

Scalability and distributed intelligence

In scalability and distributed intelligence, the three most important uses of 5 G, including industry automation, IoE, and AR/VR, are hindered by issues with speed and quality. 6G-enabled edge devices' improved performance and intelligence allow these systems to carry out complicated activities that were previously limited to elegant cloud servers. Along with resolving 5 G's problems and obstacles, 6G also satisfies the needs of industrial automation and holographic communication. The latest technique to achieve complete immersion without the need

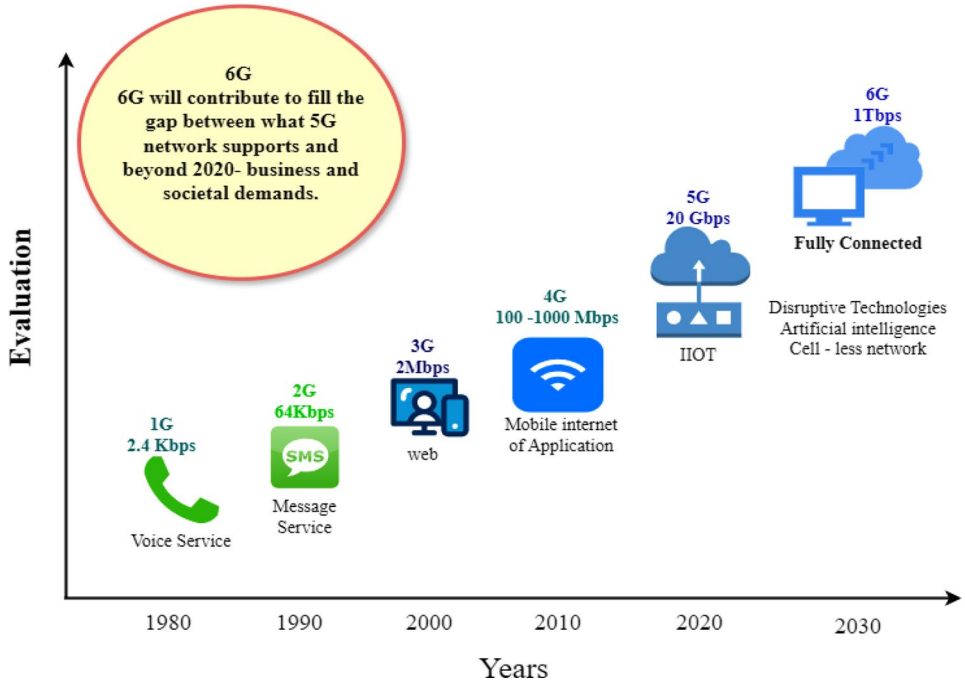


Fig. 5 Evolution of wireless communication with a timeline from 1G – 6G [10]



Fig. 6 Connectivity scenarios using 6G eURLLC [11]

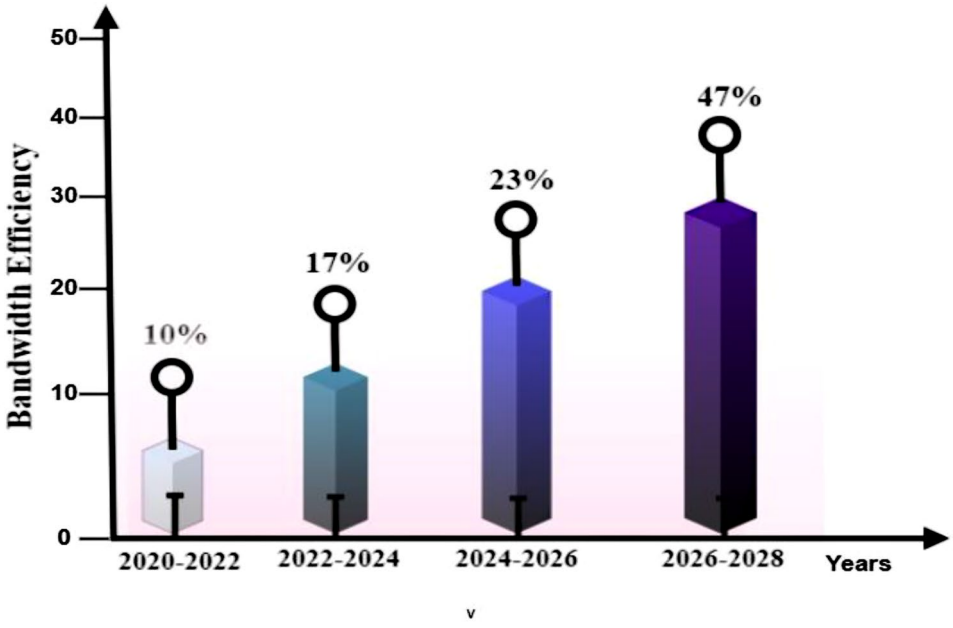


Fig. 7 Internet usage

for head-mounted devices is through holographic communication [13].

Cloud service scalability as edge devices take over specific computing workloads, 6G technologies will improve and cloud infrastructures may concentrate on increasingly complex tasks. As a result, tasks can be dynamically redistributed between the edge and cloud based on real needs and availability of resources, leading to a more distributed and more intelligent network.

Security & privacy of 6G wireless communication using fog computing

In security and privacy in the context of statistical and ML analysis, Differential Privacy (DP) is another budding privacy-preserving technology that may likely appear in future 6G wireless applications [14] and [15]. In the future, quantum computing will lead to edge, fog, and cloud computing security. The prospect for quantum security issues in 6G wireless technology is investigated by the [16] and [17].

Enhanced IOT and AI integration

The 6G network is designed to support a enormous number of edge devices, IoE, IoV, and industry automation [18]. Edge computing empowers local, real-time data processing and AI-driven decision-making at the edge, which, when combined with 6G, improves IoE and IoV applications [19]. This reduces the dependency on

cloud-based AI models for real-time analysis, enabling quicker responses and enhancing the feasibility and efficiency of applications such as linked vehicles, industries, smart healthcare systems, etc.

Energy efficiency and sustainability

Energy efficiency and sustainability are critical considerations in the design and implementation of 6G networks, particularly with the integration of edge computing. By processing data closer to the source, edge computing significantly reduces the need for data to travel long distances to centralized servers, thereby minimizing energy consumption and latency. This localized data handling not only conserves bandwidth but also enhances the overall responsiveness of the network, which is essential for real-time applications such as autonomous vehicles and smart cities. For example every daily activity in metropolitan areas, such as working and shopping, requires parking at a reasonably priced location. It is therefore possible for users to look for open parking spaces online. Finding a free parking place and inefficient parking space management are common problems with conventional parking systems. In order to address these issues, smart parking systems have been devised. These systems offer systematic parking space management as well as quick computation of available parking spaces using a variety of machine learning (ML) algorithms [20].

Moreover, edge devices can leverage renewable energy sources, further promoting sustainable practices. The synergy between energy-efficient technologies and edge computing in 6G will facilitate a more sustainable digital infrastructure, ultimately contributing to lower carbon footprints and supporting global efforts towards environmental sustainability. As networks evolve, prioritizing energy efficiency will be vital to accommodate the anticipated surge in connectivity demands while ensuring that these advancements do not come at the expense of the planet. Table 2 provides a comprehensive analysis of the comparison with recent reviews, highlighting key insights and technologies driving 6G networks, as well as unique contributions of the proposed study.

Table 2 Comparative analysis with existing approaches

Key Features	Limitations	Potential Solutions
Cloud–MEC resource	No 6G context [21, 22]	Adaptation within 6G capabilities [23, 24]
AI-driven 6G	Edge AI [25]	Ultra edge Processing, Distributed Layered Architecture
Edge & Cloud Integration	Edge Computing is addressed [26]	6G-Edge-Cloud unified architecture
Cutting Edge Technology Impact	Not Addressed [27, 28]	Quantum Computing, Blockchain, AI, FL

Impact of 6G on cloud computing

This section explores the transformative impact of EC with a 6G network on CC. To understand the impact of 6G on CC, we have proposed a process in seven phases, as illustrated in Fig. 8. The edge and cloud integrated model is tightly coupled for centralized processing and storage, providing distributed access to data through the InterPlanetary File System (IPFS) and blockchain technology, which ensures transparency and integrity. The models also utilized AI-enabled architecture and high-performance computing models to easily validate and test large language model (LLM) or agentic AI models, preserving low latency, transparency, and scalability. The framework is designed to achieve 6G's ultra-low latency and bandwidth, enabling real-time, immersive applications such as IIoT, drone integration, and the metaverse using continuous collaboration between edge and cloud infrastructures. The detailed analysis of seven phases is as follows:

Step1: Centralized processing and storage

In the realm of 6G, cloud computing will play a pivotal role as a centralized hub for processing and storing vast amounts of data. This centralized approach allows for the efficient management and allocation of computational resources, ensuring that networks can scale seamlessly to meet increasing demands. In [20], the Cache Decision System (CDS) on the Smart Buildings (SB) is studied. CDS consists of two types of servers, Cloud Server (CSe) and Edge Server (ESe), and combines the functions of the IoT with CC, EC, and big data on a smart building (SB) to operate in a wireless-mobile 6G network for managing big data (BD) on SB.

Step2: High performance computing (HPC)

As 6G networks evolve, they are expected to leverage cloud computing for high-performance computing (HPC) tasks, revolutionizing how resource-intensive applications are executed. A non-blocking metro fabric is offered by 6G to address reduced latency in the data transmission network and prevent congestion [29].

Step3: Network slicing and Virtualiztion

An upcoming method that can affordably support a number of applications is called network slicing [30]. There are three stages in the network slicing lifecycle: planning, operation, and preparation. One potential answer that 6G will offer is AI-based network slicing [31]. Data traffic loads differ over time, therefore the resource reservation needs to be flexible enough to meet changing demands in real time. Reinforcement learning (RL) techniques like deep deterministic policy gradient (DDPG) can be used to overcome this.

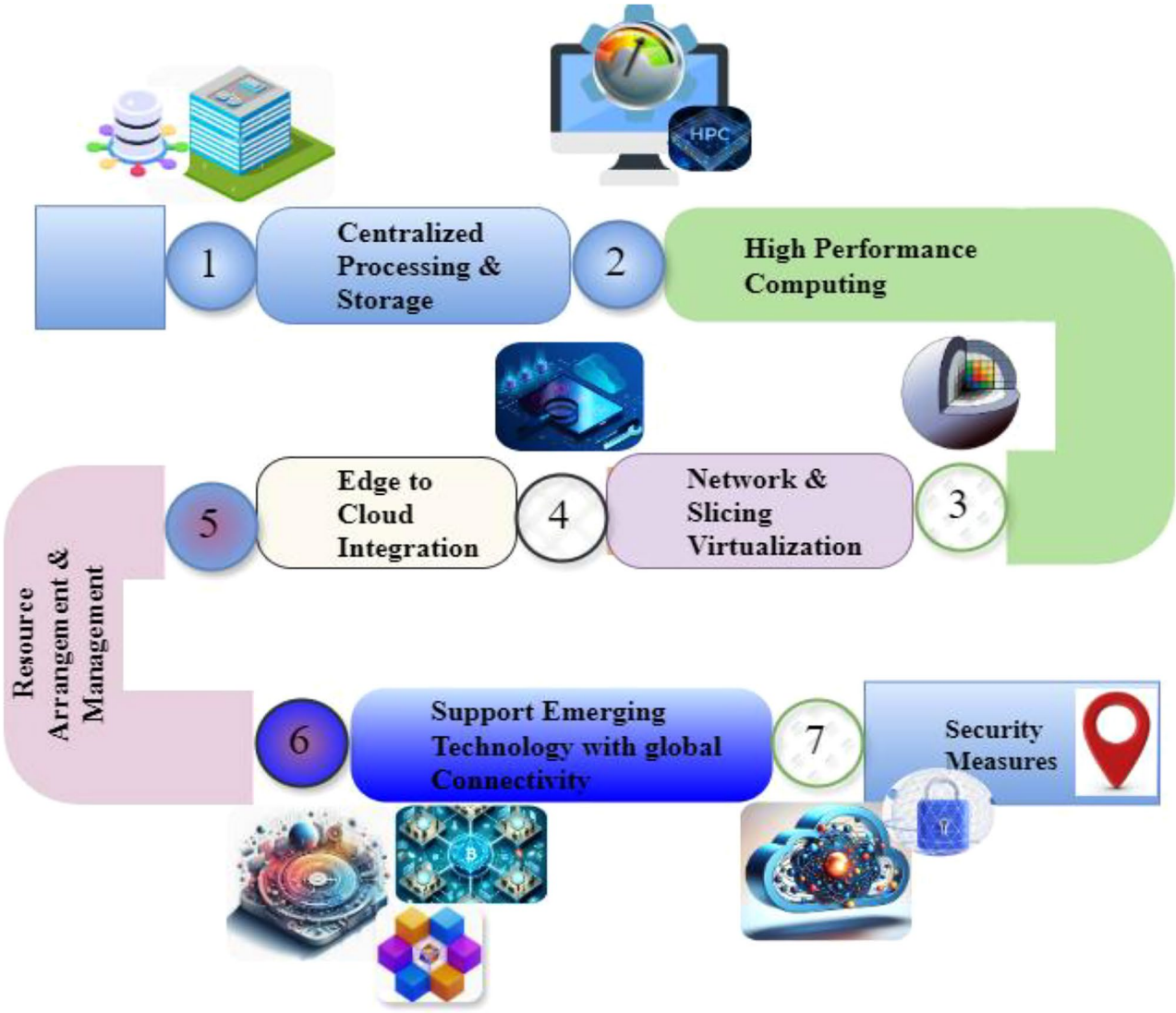


Fig. 8 Impact of edge computing with 6G on cloud computing

Cloud-based network slicing is poised to be a transformative feature of 6G, enabling the network to be segmented into virtual slices tailored to specific use cases with distinct requirements. Network slicing is a crucial factor that network operators must take into account in order to guarantee the devoted execution of virtual networks and to offer comprehensive support for the efficient delivery of services to a variety of users, including equipment, industries, and automobiles [32].

Step4: Edge-to-cloud integration

6G is set to emphasize seamless integration between cloud and edge computing, creating a cohesive ecosystem that optimizes overall network performance. The 6G networks will also provide global coverage through the effective integration of satellite and underwater communication networks [33]. Using 6G for EC has several benefits over traditional CC. EC can offer greater service performance with shorter reaction times as well as reduced service costs with less bandwidth and energy consumption by deploying cloud-like capabilities close to users. Fast learning in the large-scale 6G network is made possible by federated learning (FL), which minimizes computation and communication overhead, protects privacy without sharing datasets, and shortens client-server communication latency by utilizing computation capabilities made available by edge computing close to the user [34].

Step5: Resource arrangment and management

With the advent of 6G, cloud computing will be essential to the effective allocation and control of resources, allowing networks to adapt quickly to changing user needs. To effectively manage resources including bandwidth,

Table 3 Comparative analysis of the models and features of 5 G and 6G

Key Features	5G	6G	Impact on Edge- Cloud Architecture
Frequency Range (GHz)	24–52 GHz (mmWave)	100–300 GHz	Development of THz-compatible hardware, light weight protocols, and ultra-low-loss latency
Bandwidth (B)	100 MHz – 1 GHz (per Chanel)	10 GHz	Multi-band or wider contiguous spectrum aggregation to achieve > 1 GHz
Data Rate (R) (Gbps)	10–20 Gbps	500- to 1000 Gbps	Advanced modulation using THz MIMO to achieve upto 1000 Gbps, holographic beamforming, and AI for resources optimization
Path Loss (A(d)) (dB)	60–70 dB	A(10) ≈ 80–110 dB	Use of RIS to reduce A(d) and High-gain ultra-massive MIMO
Mobility Support	138 m/s	278 m/s	Predictive mobility management using AI and digital twins for high-speed trains & drones
Signal-to-Noise Ratio (SNR) (dB)	≈15–20 dB	SNR ≈ 30	Improved beamforming for SNR > 20, use advanced channel coding, and quantum-level error correction
Energy Efficiency (J/bit)	$10^{-9} - 10^{-7}$ J/bit	10^{-10} J/bit	AI-optimization, Efficient low-power THz circuits for 10 – 12 J/bit
Latency (T) (ms)	T < 1–10 ms	T < 1 ms	Real-time optimization for $T < 1$ ms, Compute-offloading to edge AI, and optimized MAC protocols
Edge-Cloud Architecture	Only Cloud-centric and optional edge AI enabled	Cloud-edge leveraging	Tight integration of edge computing along with decentralized orchestration.
Security	Conventional PKI Cryptosystem	Zero Trust, Blockchain and Post Quantum Cryptosystem	Post-quantum Cryptosystem, QKD and distributed lightweight authentication.

power, and processing capacity, a 6G network's resource management layer can make use of complex resource allocation algorithms [35]. Network slicing, which enables the development of several virtual networks with various features on a physical network, is expected to be supported by 6G.

Step6: Support for emerging technologies with global connectivity

In the 6G landscape, CC will be a critical enabler of global connectivity and collaboration, providing a robust platform for seamless communication and data exchange across diverse geographical boundaries. Using fast and limitless wireless connections will enable civilizations to become data-driven shortly [36]. CC will play a vital role in supporting integrating emerging technologies within 6G, including AR, VR, and advanced IoT applications. These technologies demand substantial computational resources and low latency to deliver immersive and responsive user experiences. In 6G, conventional computing systems will be replaced by quantum computing to promote parallelism, which is an inherent property of quantum mechanics. Quantum computing (QC) will enable the parallel processing of multi-dimensional and extensive-size data. It will enhance the algorithms and techniques of ML and AI that deal with a tremendous amount of data for training. QC can help provide high security in 6G networks. It can be considered the key enabler of 6G technology. It enhances network efficiency by keeping knowledge on a real-time basis and rapidly responding to user demand [37, 38].

Table 4 The abbreviations used in the study

Symbol/Indicators	Description	Use/Relevance
Bandwidth (B)	Data rate	Compare 5 G & 6G
R (Gbps)	Achievable data rate	For 5 G & 6G analysis
A(d) (dB)	Path loss at distance d	GHz/THz feasibility
(SNR) (dB)	Signal-to-Noise Ratio	Level for error correction
T (*ms*)	End-to-end latency	Edge–cloud joint optimization
E (J/bit)	Energy efficiency	Compute–communication

Step7: Security measures strengthened

In the evolving landscape of 6G, enhanced security measures rooted in CC will be essential for protecting network integrity and safeguarding user data. It is imperative to reassess conventional security approaches in light of 6G development. The more demanding requirements of future network designs must be met by advanced approaches for secure communication, access management, authentication, encryption, and the detection of malicious activity [39].

We have also provided a detailed technical analysis of the edge-cloud architecture, utilizing key features of 5 G and 6G. Table 3 summarizes the concept maps or comparative models for 5 G and 6G, highlighting the impact on edge and cloud-based systems based on their respective features.

To make the study easier to understand, we additionally provided the meanings of quantitative indicators and abbreviation, variables used in the mathematical models and comparison analysis by Table 4.

This section also explores the transformative impact of EC with a 6G network and edge-cutting technologies on CC, enabling more intricate, real-time, and data-intensive applications across multiple industries 4.0 [40]. In addition to the technological foundation of 6G networks, which encompasses a wide range of advancements aimed at achieving Edge Offloading Optimization, End-to-End Latency, Terahertz Channel Modeling, and Enhanced Conceptual Design, these areas are also explored.

End-to-end latency

The end-to-end latency of an edge node using a cloud architecture is the total time it takes for data to travel from the source edge node to the cloud node and for the result to be returned to the source edge node [41].

The end-to-end latency is decomposed as:

$$\begin{aligned}\mathcal{L}_{Total} = \mathcal{L}_{tx} + \mathcal{L}_{prop} + \mathcal{L}_{queue} \\ + \mathcal{L}_{proc} + \mathcal{L}_{backhaul} + \mathcal{L}_{retrans}\end{aligned} \tag{1}$$

$$\mathcal{L}_{tx} = \frac{S}{R} = \frac{Datasize\,(bits)}{Acheiveddatarate} \tag{2}$$

$$R = B\log_2\left(1+\gamma\right) \tag{3}$$

Here $\mathcal{L}_{tx}$ is a transmission latency. γ is instantaneous SNR.

In 6G Communication achieved data rate will considered as 1TB.

$$\mathcal{L}_{prop} = \frac{d}{v} \tag{4}$$

$$\mathcal{L}_{prop} = \frac{(Distance\ between\ edge\ nodes)}{Speed\ of\ light\left(3\times10^8 m/s\right)}$$

In 6G THz links, so propagation delay is considered as extremely low (<0.1 *ms*).

$\mathcal{L}_{queue}$ is a queuing delay with stochastic modeling

$$\mathcal{L}_{queue}^{M/M/1} = \frac{\boldsymbol{\rho}}{\boldsymbol{\pi}(1-\boldsymbol{\rho})},\ \rho = \frac{\boldsymbol{\lambda}}{\boldsymbol{\rho}} \tag{5}$$

Queue type is M/M/1, Markovian arrival rate, Markovian service rate, and one server for queue handling, respectively.

ρ is traffic intensity or traffic load on the system ($0 < \rho < 1$).

For bursty traffic:

$$\mathcal{L}_{queue}^{M/G/1} = \frac{\lambda E[S]^2}{2\left(1-\rho\right)} \tag{6}$$

Queue type is M/G/1, Poisson arrivals, General service distribution rate, and one server for queue handling, respectively.

λ is arrival rate i.e. task per second, S is random variable for service time, and $E[S]^2$ moment of service time i.e. variance & mean^2,and ρ is server utilization.

$$\mathcal{L}_{proc} = \frac{C}{f_{edge}} \tag{7}$$

Where C is CPU cycles and f_{edge} is edge processor frequency.

$\mathcal{L}_{backhaul}$ is the backhaul latency, i.e., the time required for data to travel from edge node to cloud node. In 6G, ultra-high-capacity THz and optical wireless backhaul drastically reduces this latency. Similarly, $\mathcal{L}_{retrans}$. is the retransmission latency. In 6G URLLC requires:

$$Pr\left(L_{total} \le 1ms\right) \ge 1 - 10^{-5}. \tag{8}$$

Channel modeling

Channel Modeling in 6G: Channel Characteristics in the THz Band [42]. It differs from those of lower frequency bands (mmWave and cmWave).

In Terahertz Channel Modeling, THz Path Loss is

$$PL_{THz}\left(d,f\right) = 20\log_{10}\left(\frac{4\pi f d}{c}\right) + \alpha\left(f\right)d \tag{9}$$

Here $\alpha\left(f\right)$ models molecular absorption.

$20\log_{10}\left(\frac{4\pi f d}{c}\right)$ is free space path loss, f is frequency, and d is distance

Molecular Absorption Noise is

$$N_{abs}\left(f\right) = k_B T_0 \alpha\left(f\right) B_{THz} \tag{10}$$

Here k_B is Boltzmann constant, $T_0$is standard temperature, and $B_{THz}$is channel bandwidth.

Beamforming Gain is

$$G_{BF} = N_{ant} \tag{11}$$

Here N_{ant} is number of antenna elements.

$$\gamma THz = \frac{P_t G_{BF}}{PL_{THz}\left(d,f\right)\left(N_0 + N_{abs}\right)} \tag{12}$$

Here, $P_t$is transmission power, is G_{BF}Beamforming gain raises received power, $PL_{THz}\left(d,f\right)$is High attenuation,

N_0 is thermal noise, and $N_{abs}$is extra molecular absorption noise.

Edge offloading optimization

To minimize latency and energy consumption, offloading [43] decision $\beta \in \{0, 1\}$ is chosen as:

$$\min_{\beta} \mathcal{L}_{total}(\beta) + \lambda_e E(\beta) \tag{13}$$

Subject to URLLC constraint:

$$\mathcal{L}_{total}(\beta) < 1ms \tag{14}$$

Energy consumption:

$$E(\beta) = \begin{cases} P_{CPU} \frac{c}{f_{local}} & ,\beta = 0 \\ P_{tx} \mathcal{L}_{tx} & ,\beta = 1 \end{cases} \tag{15}$$

Here, $\mathcal{L}_{total}$is total latency, $\beta = 0$ and $\beta = 1$ denotes local execution and offloaded execution, respectively.

Conceptual diagram

The End-to-End (E2E) communication and computation in a 6G network enabled edge to cloud computing layered architecture is illustrated in Fig. 9. It is a clear representation of data packet travel from an end user device (IoT/UE) in a 6G network using Edge Compute Node, a Cloud Server node, and vice versa. Each arrow represents a delay component that contributes to the total end-to-end latency from the edge node to the cloud node.

6G enabled use cases & key enabled technology for Edge/Cloud computing

This section explores the transformative impact of EC with a 6G network and edge-cutting technologies on CC, enabling more intricate, real-time, and data-intensive applications across multiple industries 4.0 [40].

Holographic communication and immersive experiences

Holography is a technology that is used to record and generate holograms. The holograms are generally known as the original properties of depth and parallax of not only 3D humans but also objects or items [44]. The Holography was introduced by the digital market. The science of creating holograms, or holography, is founded on the interference principle. In addition to enabling real-time interaction with 3D holograms, 6G will provide ultra-immersive virtual & augmented reality (VR & AR) experiences for learning, amusement, and distant collaboration [45]. 6G distribution allows for network caching in AR and VR. The names AR and VR are two instances of how technology has made it harder to distinguish between the two [46]. To minimize latency, edge computing shall handle real-time data processing and holographic and immersive content rendering. CC will enable complex computational processes and large-scale data storage at the same time.

Massive iot and smart environments

The Internet of Things (IoT) is a computing and communications paradigm whereby everyday objects are connected to the Internet [47]. Therefore, the IoT is a fundamental enabler of smart environments, such as smart homes, smart health, smart cities and factories, etc. Examples of such services can be those concerning the security and safety of inhabitants (for example, the anti-intrusion alarms or the detection of gas leaks) [48, 49]. While cloud computing will manage data storage, machine learning model training, and worldwide coordination for IoT ecosystems, edge computing will handle local data processing and real-time decision-making [50].

Enhanced mobile broadband (EMBB) and ultra-reliable low-latency Communications (URLLC)

Three usage scenarios are defined by the International Telecommunication Union (ITU): massive machine type

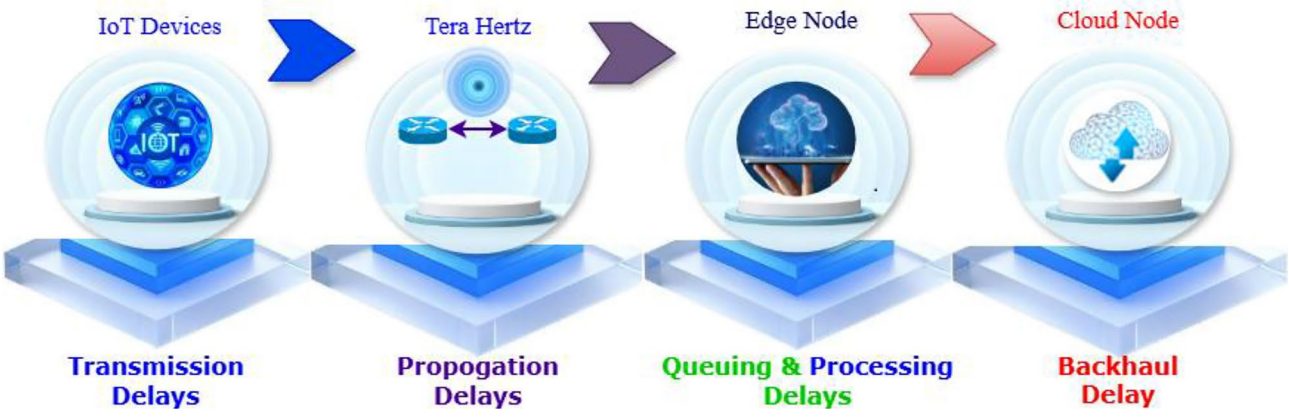


Fig. 9 Conceptual diagram of delays in 6G

communication (mMTC), enhanced mobile broadband (eMBB), and ultra-reliable low latency communication (URLLC) [51–55]. To maintain or meet the competitive criteria for the applications of wireless networks, industry and academia have started to conceptualize the sixth generation (6G) of wireless communication systems [56] - [57]. For mission-critical applications like emergency response and autonomous driving, 6G will provide two benefits: enhanced mobile broadband and ultra-reliable low-latency communications. With 6G, high-speed access will also be feasible in crowded locations.

Artificial intelligence of Things (AIOT)

The linked artificial intelligence (AI) will be expected to be the major emphasis and a critical aspect of 6G [58]. 6G will enable the integration of AI with IoT, enabling intelligent networks to process enormous amounts of sensor data in real time using AI algorithms for predictive maintenance, smart logistics, and environmental monitoring. Edge AI provides an efficient framework for linked intelligence by streamlining data gathering, processing, transmission, and consumption at the network edge [59]. 6G will have a significant impact on edge computing since edge devices will perform inference tasks locally, allowing for real-time replies and reducing the need for data transfer, while AI models will be formed in the cloud to address all computing challenges.

Digital twin technology

A virtual model of real physical things is designated as a digital twin (DT) technology. During the object's lifecycle, it simulates the behaviour and keeps an eye on operations using real-time data served by the object's sensors. DT technology was originally devised for industrial and manufacturing applications [60]. Fig. 10 shows all the industry applications of DT-enabled technology. A digital twin network (DTN) is applicable in modelling a group of physical entities with complex interactions among themselves. Thus, a DTN is an extension of the DT concept by having multiple DTs communicating with each other [62]. 6G will improve DT technology by enabling real-time synchronization of digital and physical entities, including cities, buildings, and smart factories. This

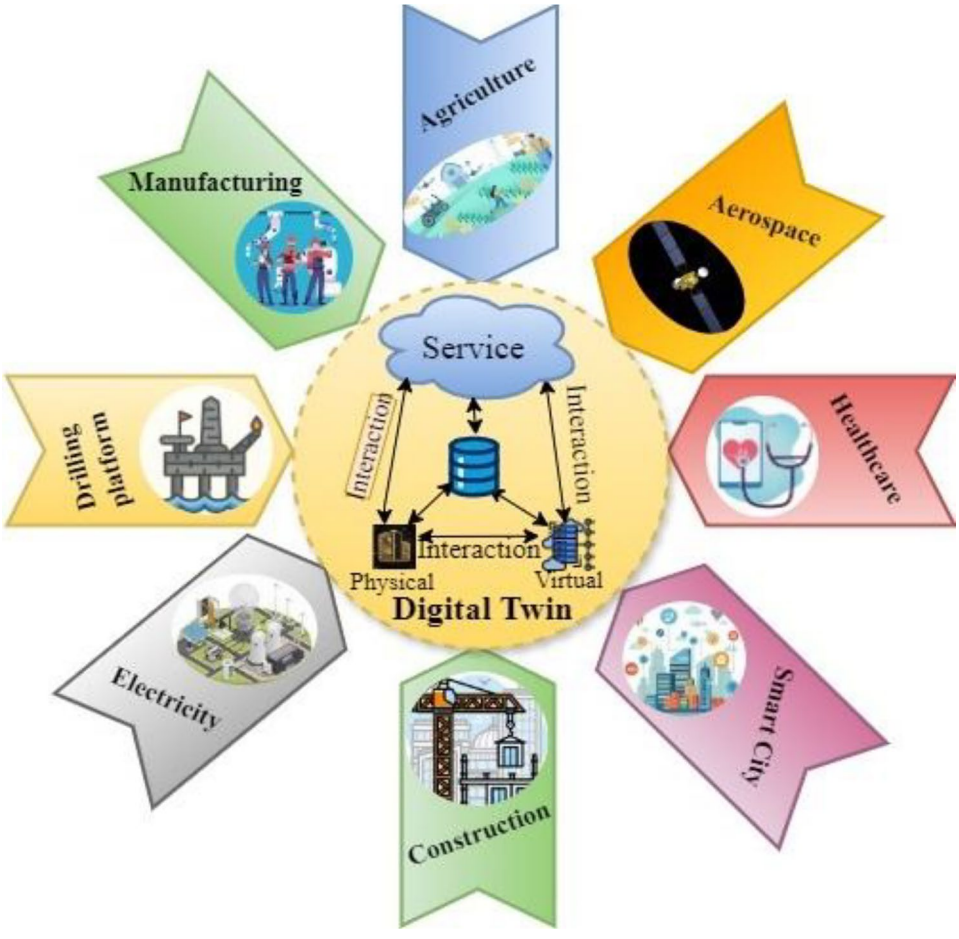


Fig. 10 Impact of edge computing with 6G on cloud computing [61]

will allow for simulation, proactive maintenance, and optimization.

Augmented reality & virtual reality

Ubiquitous connectivity will be made possible *via* 6G, and MR, VR, and AR experiences will be seamlessly integrated with Metaverse. The term "metaverse" is a combination of the terms "verse," which is the simplified form of the universe, and "meta," which implies beyond [63]. Real-time interaction between VR and AR applications and the physical world is made possible by 6G.

Figure 11 illustrates the AR/VR integration. As edge computing processes data closer to the user, it minimizes latency and allows real-time rendering for AR/VR applications. For interactive metaverse activities like multiplayer gaming or virtual collaboration, this guarantees responsive, efficient experiences.

Blockchain technology & metaverse

The next generation of technology that will transform our real-world existence into a virtual one is the metaverse, non-fungible tokens, and blockchain [64, 65]. Blockchain is a decentralized technology that makes it possible to record digital transactions transparently and safely [66]. Using blockchain technology, the non-fungible token (NFT) acts as a certificate of ownership and proves who owns digital assets [67]. Combining blockchain technology with 6G networks has several advantages and potential uses, as shown in Fig. 12. With its anticipated improvements in latency, higher network capacity, and data transfer rates, 6G can handle complicated transactions and applications in real-time.

Enhancing communication and transaction security and privacy is one possible application of blockchain technology in 6G [68]. The higher throughput and speed of 6G networks may allow blockchain-based solutions to improve data security and defend against hacks and breaches. Creating decentralized networks and platforms through integrating blockchain technology with 6G is another use. Developing decentralized networks that safely store and handle vast volumes of sensitive data, including financial transactions or medical information, may be made possible by 6G's increased speed and capacity. Compared to centralized systems, these decentralized solutions provide better security and privacy and encourage the creation of new services and applications. Centralized cloud computing has become complex in fulfilling several quality-of-service (QoS) demands of different applications due to the drastic rise in IIoT/IoT devices [69].

Federated learning

By storing data locally and only exchanging model updates with the cloud, FL provides privacy benefits, as shown in Fig. 13. FL will be able to manage more sensitive applications, like healthcare, finance, and personal data, without running the risk of data breaches because to the higher bandwidth and security characteristics of 6G [7,

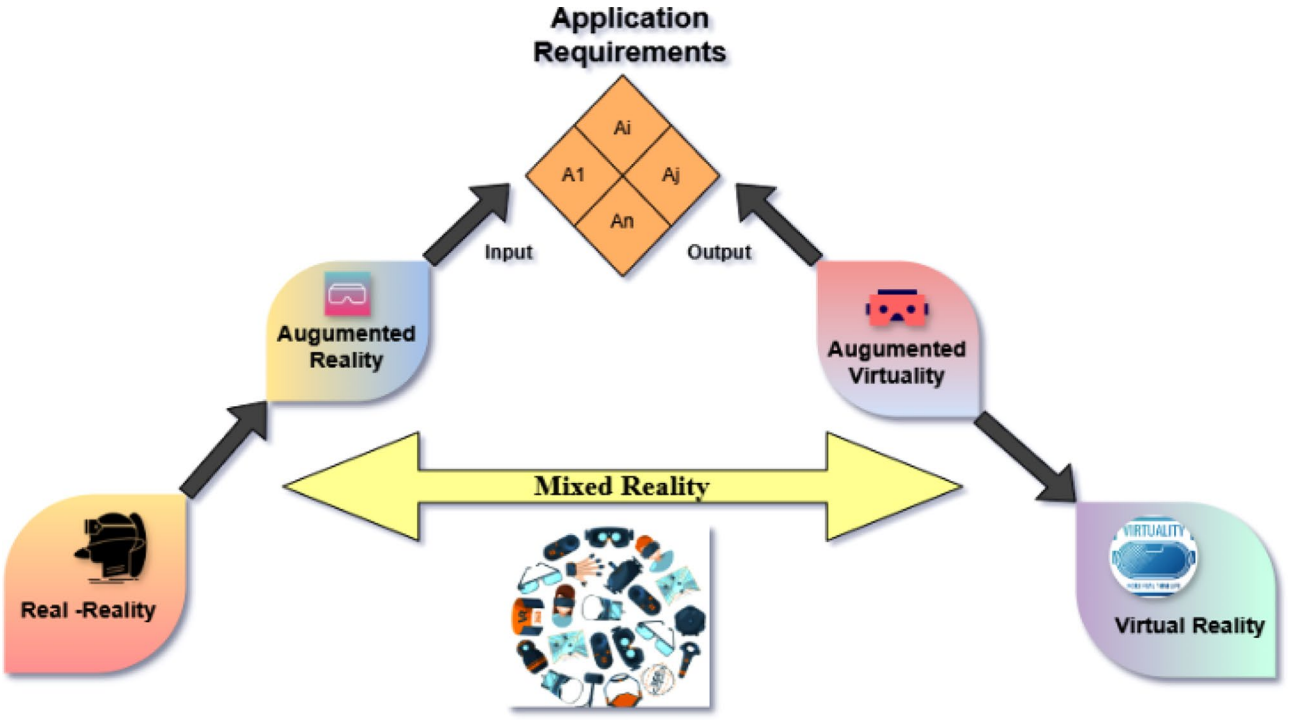


Fig. 11 AR/VR integration



Fig. 12 Integrated technologies of metaverse [21]

71, 72]. With the development of deep learning technologies, the 6G communication system has spawned many emerging infrastructures such as Integrated Terrestrial and Space, FL networks, decentralized infrastructures, and trustable infrastructure.

Network slicing

Network slicing will be significantly transformed by the interaction of edge computing with 6G networks, enabling the creation of dynamic, effective, and customized virtual networks that can meet a wide range of application requirements. The process of dividing physical network infrastructure resources into distinct logical networks is known as "network slicing," making it easier to divide up networks, services, and applications vertically [73]. Network slicing with 6G will enhance the flexibility and scalability of CC & EC with reliable and secure services for a range of use cases and industries.

Quantum computing

Quantum computing in 6G is poised to revolutionize network capabilities by enhancing processing power and data security, particularly when integrated with edge and cloud computing [74]. Sixth-generation (6G) conceptualizes an evolutionary communication platform relying on network software, satellite, and ultra-dense networks that need ultra-high speeds, tactile response time, and a cost-effective network. Quantum computing has been

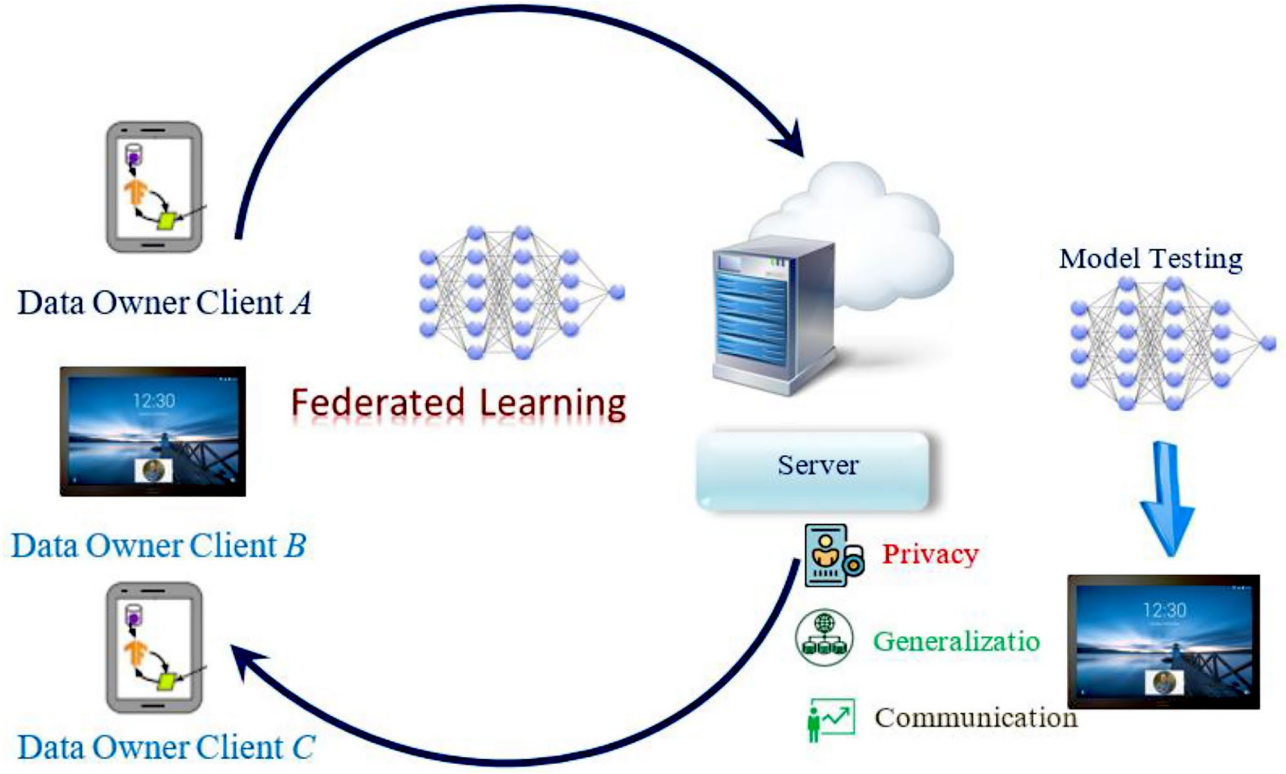


Fig. 13 Federated learning model [70]

considered to exploit the powerful and robust quantum mechanics to support fast computing tasks [75, 76]. Recently, quantum computing has been applied to support real-time optimal resource allocation in 6G networks. Fig. 14 illustrates the real applications and future direction of quantum computing with 6G, edge, and cloud computing.

AI-enabled zero-touch self-organizing networks in 6G integrating edge-cloud computing

5G relies on predefined protocols with human intervention for optimization; however, 6G demands highly autonomous and intelligent solutions to manage the network resources efficiently. The proposed framework tightly couples AI to allocate bandwidth, optimize spectrum dynamically, and improve real-time security [77]. The framework is used for smart city applications, including EC, FL, Digital Twin (DT), and AI to enable full automation in 6G networks. In this context, the framework used Zero-touch Self-Organizing Networks (ZSON) enabled by DT and Deep Reinforcement Learning (DTDRL-ZSON) for dynamic network slicing and resource provisioning in 6G edge networks.

The overview of the DTDRL-ZSON framework [77] is presented in Fig. 15. The framework manages slices, predicts latency, forecasts traffic, evaluates slicing decisions, and provides synthetic feedback for faster learning. It also uses Deep Deterministic Policy Gradient (DDPG) for continuous resource provisioning and latency prediction in 6G networks with CC & EC. The DTDRL-ZSON framework is evaluated against EdgeSlice and TARO, demonstrating superior performance. Furthermore, an algorithm for the same framework is also explored.

The DTDRL-ZSON algorithm is used federated average ($FedAvg$) concept where the global model parameters are updated as θ

$$\theta \leftarrow \theta + \eta \sum_{i} w_i.\theta_i \tag{16}$$

Here, learning rate and weight is η and w_i, respectively. The resource allocation will be real-time and unlike 5G based on fixed constraints. For easy understanding of the algorithm process, the meanings of the acronyms are defined in Table 5.

$FedAvg$ is based on local training at each edge node, it selects an optimal strategy for network slicing and resource allocation

$$a_i = \pi\left(\theta_{\mathbf{i}}, s_{\mathbf{i}}\right) \tag{17}$$

Here, $\theta_i$will be updated based on local parameters (s_i, a_i,r_i,s_i') to fulfill Service Level Agreement (SLA)

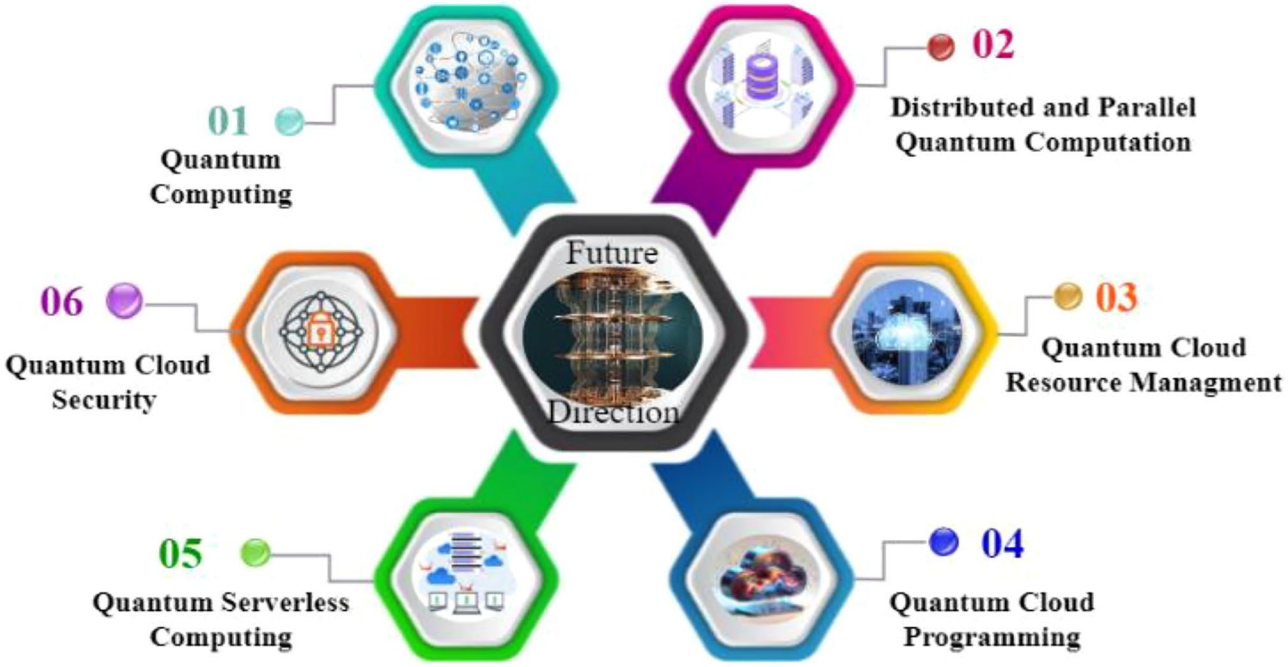


Fig. 14 Future directions on quantum computing with CC & EC [74]

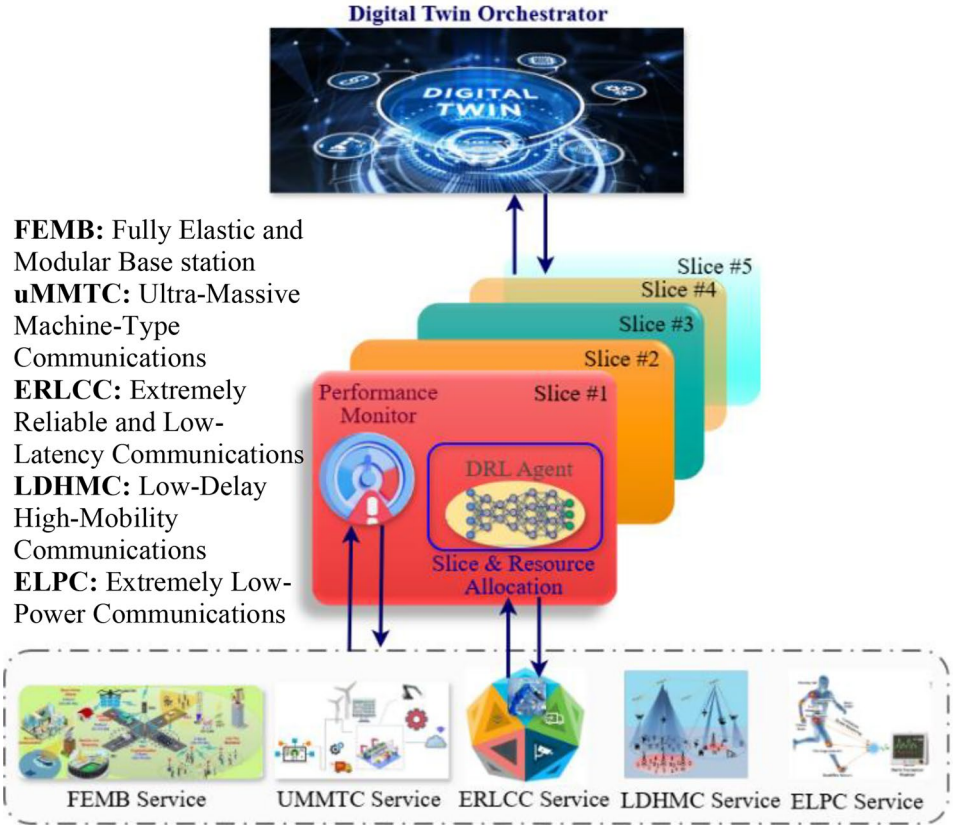


Fig. 15 The DTDRL-ZSON framework [77]

Algorithm: **DTDRL-ZSON**

Input: Edge devices: $\mathcal{N}$, DTO at Cloud, $DDPG$ parameter: θ, Local Training Rounds: $\mathcal{T}$, Communication Rounds: $\mathcal{R}$, State Space: $\mathcal{S}$, Action space: $\mathcal{A}$, Reward function: $R(s,a)$
Output: Optimized global $DDPG$ policy $\pi^*(\Theta)$ for E2E slice management and resource provisioning
Data: $\mathcal{D}$

Initialize: Global $DDPG$ policy $\pi(\Theta)$ with random weights Θ and distribute Θ to all edge devices $i = 1$ to $\mathcal{N}$
foreach communication round $r = 1, \ldots\ldots, \mathcal{R}$ **do**
  foreach each edge device i in $\mathcal{N}$ **do**
    Observe local state s_i using $D\mathcal{T}_i$:
    s_i= QoS requirement, resource availability and $\mathcal{L}$
    for local training step $t = 1$ to $\mathcal{T}$ **do**
      Selection action $a_i = \pi(\theta_i,\ s_i)$
      Apply a_i to perform slicing and resource allocation
      Receive reward $r_i = \mathrm{R}(s_i, a_i)$
      Observe new state s_i'. Update local parameters θ_i using (s_i, a_i, r_i, s_i')
    Send model update $\Delta\theta_i$ to DTO
  ***DTO* executes:**
  Aggregate Updates $\theta \leftarrow \theta + \eta \sum_i w_i.\theta_i$
Broadcast updates to all $\mathcal{N}$ (edge devices)

Table 5 List of common acronyms and Description used in the paper

S. No.	Acronym	Meaning
1	$\mathcal{N}$	Edge Devices
2	DTO	Digital Twin Orchestrator
3	$DDPG$	Deep Deterministic Policy Gradient
4	θ	Global $DDPG$parameter
5	$\mathcal{T}$	Local Training Rounds
6	$\mathcal{R}$	Communication Rounds
7	$\mathcal{S}$	State Space
8	$\mathcal{A}$	Action space
9	$R(s,a)$	Reward function
10	$\pi^*(\Theta)$	State of the art
11	$\mathcal{D}$	Dataset
12	$\mathcal{L}$	Latency

requirements and optimizing resource utilization and minimizing a predefined loss function.

Vision of edge computing with 6G network on cloud computing

A highly flexible, intelligent, and efficient ecosystem that blends the most significant aspects of cloud and edge computing is the vision for EC with 6G networks in CC. Cloud data centres that handle large volumes of data will have less work to do since fog and edge computing will allow data to be processed locally at the edge. EC is a technology that enables edge devices to perform computation and storage operations closer to the edge [78]. Fog computing will be a vital component of 6G-enabled massive IoT. By installing several fog nodes at various locations, devices can offload their tasks and cache data onto these nodes. Fog computing in 6G is a vital intermediate layer between cloud and edge computing, enhancing the overall network architecture by providing localized processing and storage capabilities.

The vision of 6G is to help distribute cloud infrastructure among edge nodes (micro data acquisition), resulting in a multi-computing or cloud environment that is more reliable and effective (macro data acquisition) as shown in Fig. 16.

Edge AI, TinyML, and Advanced Radio Access Technologies (RAT) in 6G are pivotal in shaping the next generation of wireless communication by facilitating higher data rates, lower latency, and improved connectivity [79]. Edge AI and tiny ML are set to revolutionize 6G by enabling intelligent processing at the network's edge, allowing for real-time data analysis and decision-making with minimal latency.

At first glance, the main vision of the 6G technology aims to provide even faster data speeds, lower latency, greater capacity, and more reliable connections than its predecessor [79, 80]. The TinyML paradigm emerges as a game-changer when considering the constrained resources of edge devices for executing artificial intelligence algorithms, as also predicted in the vision of 6G architecture.

By extending the network's capabilities beyond the edge, fog radio access networks (F-RANs) promote collaborative approaches and minimize time [81]. F-RAN plays the interface among centralized unit (CU) and user equipment (UE). Advanced Radio Access Technologies (RAT) in 6G are pivotal in shaping the next generation of wireless communication by facilitating higher data rates, lower latency, and improved connectivity. The cloud-based random access network (RAN) was initially proposed to effectively reduce capital and operational expenditure [82]. The RATs connect to the cloud, and their demands for computing and caching are served by the cloud in a centralized manner. The computational holographic radio extracts available information from the received interference to achieve precise control of the whole communication environment [83].

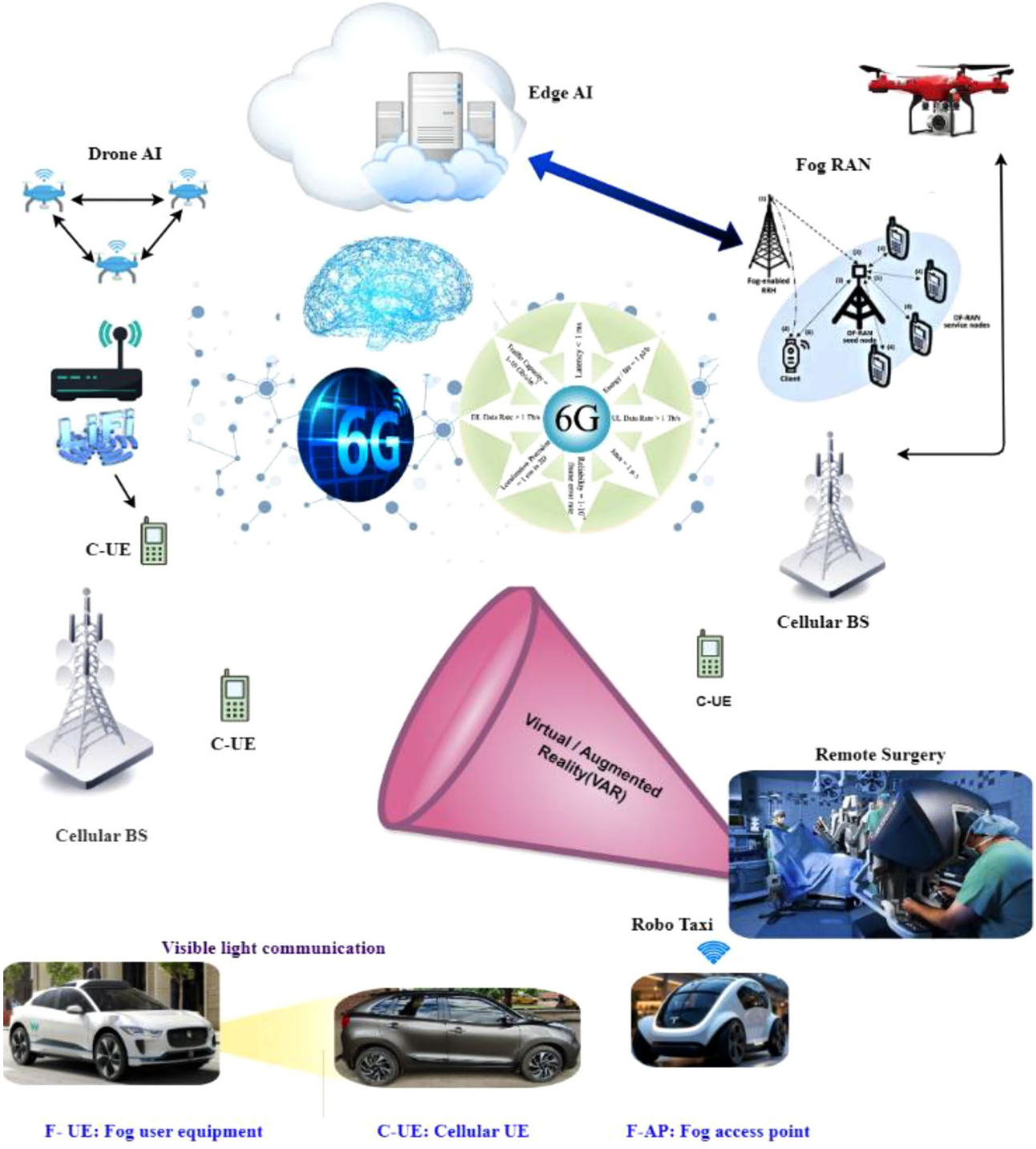


Fig. 16 The vision of 6G [8]

Futurestics 6G applications

Some effective and real-life applications are listed below to help understand the potential of 6G enabled edge-fog-cloud computing. This section extensively discusses the futuristic application of 6G-enabled edge computing, as illustrated by Fig. 17.

Smart cities

Future technological advancements like 6G and smart cities have the potential to fundamentally change our way of living and working [84]. In order to achieve intelligent services enabling autonomous network improvement, knowledge discovery, and smart resource optimization, a novel 6G intelligent AI-enabled architecture has been



Fig. 17 Futuristic 6G applications

presented in. Smart cities enabled by 6G networks, which are transforming our way of living, working, and interacting, are a symbol of the next generation of urban innovation [85]. As an outcome, smart cities develop into thriving communities that prioritize quality of life and environmental protection in addition to becoming hubs for technology.

Industrial automation

The 6G is applied to new communications networks that have evolved throughout the past few years and have incorporated various technologies, such as sensitive sensors, autonomous vehicles, immersive media, IoT, and IIoT. In 6G, the information between machines and robots will be provided in partial time units to support various operations safely [86]. The logic of EC is that the computing should be done in the vicinity of data sources. In addition, fog computing can replace edge computing. Applications and tools of cloud edge automation enable cloud computing services to provide the ability to adjust traffic rates quickly and fall automatically, which leads to reduced operational costs and reduced time needed to generate revenue.

Agricultural & environmental monitoring

Agriculture is the foundation of human society. It plays a vital role in maintaining social stability and promoting other industries. The concept of smart agriculture, which was initiated in 2017, is defined as the deep integration of modern science and technology with agriculture

so that agricultural cultivation and production can be unmanned, automated, and intelligent [87]. It will drive a comprehensive digital transformation of all industries, integrating the physical, biological, and digital worlds. As such, it is foreseeable that 6G technology will have the potential to drive the development of smart agriculture even further. Sensors and drones could monitor wildlife habitats, track animal movements, and analyze environmental data. EC and 6G will support drones and IoT devices in agriculture, allowing farmers to monitor soil conditions, crop health, and weather patterns in real-time [88]. This enables precision irrigation, fertilization, and pest control, maximizing crop yield and resource efficiency.

Healthcare sector & telemeditions

The 6G network not only supports the services provided in the previous generations but also endorses the new services by including new technology like high-level digital twinning, Embedded systems everywhere, multi-sensor data fusion to create multi-verse maps, and Autonomous systems. The most critical demand for 6G wireless networks is the ability to handle massive volumes of data at a very high data rate per device. The transformation to 6G technology has greatly improved the biomedical and healthcare sector. The technology is expected to revolutionize the sector from remote activity surveillance to remote robotic surgery in the most efficient way [89]. 6G robotics may create remote surgery so distant physicians can manage the procedure using robotic devices with millisecond latency and great dependability. 6G and edge computing would ensure data is processed near the surgical site, minimizing latency and enabling precise, remote control of surgical instruments.

Extended reality (XR) & metaverse

Extended Reality (XR) technologies consist of a group of innovative and immersive technologies such as AR, VR, and MR, which bring physical objects into digital environments and take digital objects into physical reality [90, 91]. The metaverse can be considered to have originated from Neil Stephenson's science fiction novel Snow Crash, published in 1992 [92]. Matthew Ball provided a comprehensive definition of the metaverse in [93]. Specifically, the metaverse is a persistent, interconnected, and large-scale 3D virtual world network, which can provide immersive experiences for numerous users simultaneously while supporting the mapping of physical worlds to virtual worlds and can also change human life and the operational mode of the industry.

Trading and financial services

The 6G is expected to support AI fully, enhanced mobile broadband, massive IoT, cell-free communications, holographic beamforming, and high data analytics. The 6G needs to increase capacity, not only in terms of bandwidth but also in terms of coverage. The multi-party and multi-dimensional real-time resource allocation will become a new typical service in the cyber twin-based 6G network [94]. 6G's low latency will allow for real-time processing of financial data at the edge, providing faster transaction speeds and reducing delays in automated trading platforms. Integrating reconfigurable intelligent surfaces, terahertz communications, and AI in the realm of 6G exploration is particularly compelling [95]. The primary goal of 6G is to address the evolving demands of the information society by 2030, envisioning a future where 5 G's limitations are surpassed, and unmet service needs are addressed. The four main components of 6G's vision are Intelligent Connectivity, Deep Connectivity, Holographic Connectivity, and Ubiquitous Connectivity, shaping the overall 6G vision [96]. Edge computing can enable real-time analysis of user data, allowing financial advisors or AI-driven applications to offer personalized advice based on up-to-the-minute financial insights tailored to individual investment goals and risk profiles.

Edge centric technological open challenges, implementation analysis and discussion

This section highlights the technological and implementation challenges of an edge-cloud-centric framework using 6G.

Technical challenges

The technical challenges encompass a broad range of important factors essential to accomplishing the ambitious objectives of 6G. Figure 18 illustrates the key challenges for 6G networks, encompassing Terahertz (THz) Communication & Propagation Challenges, Edge–Cloud Constrained Networking, Reconfigurable Intelligent Surfaces (RIS) Design & Control, AI and Autonomous Networking, Communication Spectrum & Interface Management, Energy Efficient & environmental sustainability impact, and Quantum Technology Integration.

- Terahertz (THz) Communication & Propagation Challenges

The THz band communicates beamforming enables high-speed wireless in 6G networks [97] to address the challenges in the high-frequency band and path loss. The other significant challenges are beam misalignment in high mobility, severe path loss even for short distances, blockage by physical barriers, ultra-precise beam tracking, and molecular absorption, including buildings and the human body [98].

- Edge-Cloud Constrained Networking



Fig. 18 Technical challenges in 6G networks

Edge-cloud computing provides computing at the proximity of data sources [99]. 6G networks aim to support massive data rates, large language models, and AI-driven services along with ultra-low latency. The edge-cloud layered architecture can achieve all addressed objectives with 6G; however, some technical constraints hinder end-to-end performance [100]. Dynamic and heterogeneous edge nodes and their topologies, scalability, massive connectivity, and security across distributed edges are significant challenges.

- Reconfigurable Intelligent Surfaces (RIS) Design & Control

RIS has emerged as a core enabler technology in the ever-evolving landscape of 6G networks [101]. It offers a compelling solution by deliberately manipulating the radio propagation environment; however, practical implementation and control of RIS incur several significant challenges, such as signal processing, estimation of extremely high-dimensional channels, scalability, security, and privacy [102].

- AI & Autonomous Networking

AI integration across the end-to-end edge-cloud infrastructure with 6G is evolving, optimizing, and autonomous, enabling self-evolving, self-optimizing, and autonomous networking [103–105]. Integration faces several challenges, including managing massive supervised and unsupervised data, ensuring transparency and explainability, integrating multi-agent systems into AI, addressing security vulnerabilities, and establishing trust.

- Communication Spectrum and Interference Management

In 6G, a heterogeneous spectrum architecture ranging from sub-6GHz to mmWave (30–100 GHz), Terahertz (THz: 0.1–10 THz), and ultra-wide spectrum introduces unprecedented interference patterns, resource allocation complexity, and coexistence challenges across an edge-cloud layered architecture [106]. In addition, quantum noise, RF noise, cognitive sensing, and beam collisions are also significant challenges that have yet to meet the demands of ultra-high spectral efficiency.

- Energy Efficient & Environmental Impact

Sustainable and effective solutions are required since 6G infrastructure's energy and cloud-edge collaborative environment requirements could greatly increase the carbon footprint [107]. Even while 5 G has improved energy efficiency, it still doesn't meet the environmental requirements for 6G. These challenges require accurate classification of various workloads and the provision of low-latency, efficient processing while reducing energy inefficiencies and task failures.

- Quantum Technology Integration

Integrating quantum technology into quantum communication, computing, and sensing within a unified

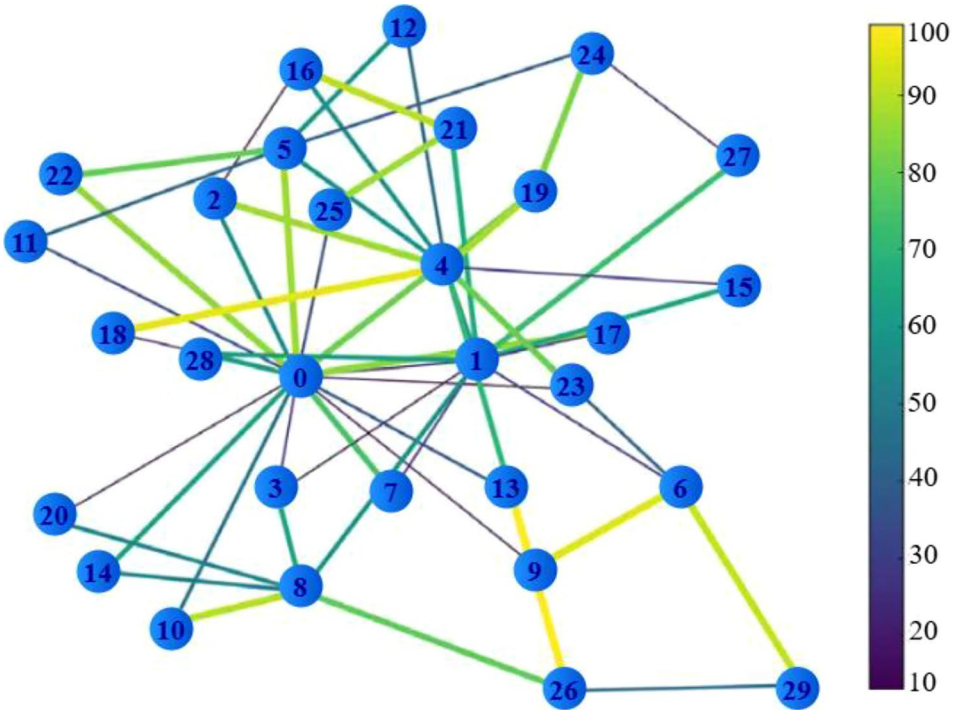


Fig. 19 Network traffic topology

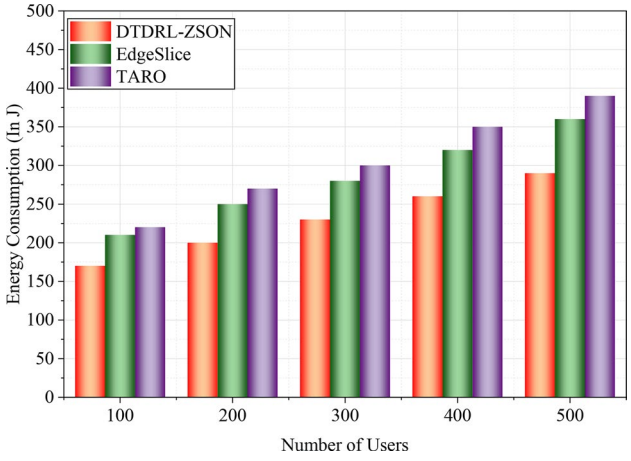


Fig. 20 Energy consumption using DTDRL-ZSON

edge–cloud computing fabric is also a top-priority objective of 6G technology. Qubit scalability at edge node, resource scheduling, edge scalability, quantum Blockchain [108], and entanglement distribution across dynamic 6G networks are also significant challenges of simulation and practical deployment.

All of these challenges constitute an extensive road map for achieving the 6G objective, blending technology innovation with innovative, flexible, and adaptable approaches.

Emptical results

This section present the validation of the DTDRL-ZSON framework and followed by highlights the technological and implementation challenges of an edge-cloud-centric framework using 6G. To validate the DTDRL-ZSON framework, a comprehensive simulation was conducted using OMNeT++, which emulates a 6G edge environment under dynamic and heterogeneous network conditions. The simulation considers 500 mobile devices generating latency-sensitive service requests to 5 edge servers within a power-law network topology comprising 30 nodes [77]. The simulation environment is illustrated in Fig. 19.

Figures 20 and 21 prove the efficacy of the DTDRL-ZSON framework [77] over EdgeSlice and TARO across network loads up to 500 users, energy consumption, and average latency, respectively.

As shown in Fig. 20, DTDRL-ZSON consistently has the lowest energy consumed across all loads compared to EdgeSlice and the TARO scheme. At peak load (500 users), its consumption remains below 290 Jules due to resource-aware task allocation and edge device selection that reduce redundant computation and data transfer. The average latency is below 15 *ms*, further reinforcing the efficiency of DTDRL-ZSON, in Fig. 21. However, the other scheme consumed more than 6–7 *ms*. The empirical results validate that DTDRL-ZSON outperforms baselines and scales well for dynamic, heterogeneous 6G edge networks.

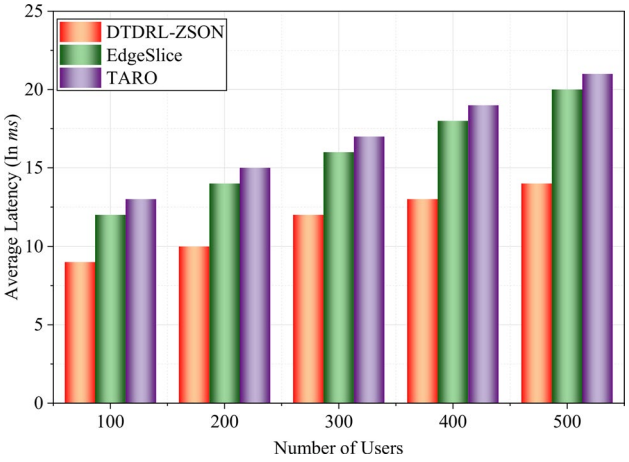


Fig. 21 Average latency analysis using DTDRL-ZSON

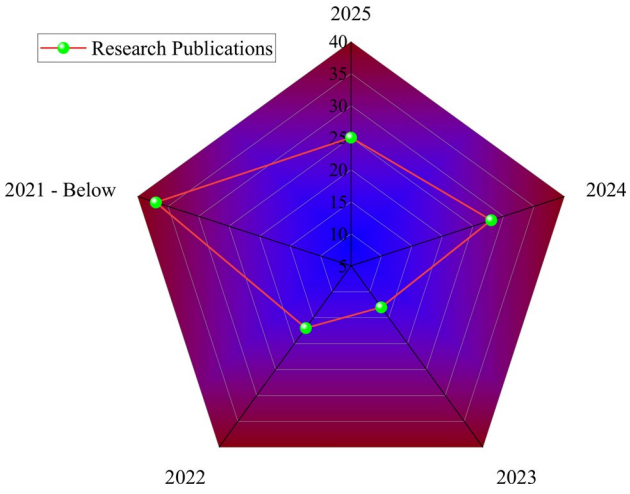


Fig. 23 Research studies annually analysis

We have also analyzed the latency communication delay [109–111] using 5G and 6G, illustrated by Fig. 22. The transmission delay using 5G and 6G is up to 2 *ms* and 0.5 *ms*, respectively, due to the use of sub-6GHz and mmWave by 5G technology, as well as ultra-wide THz/μWave bandwidths, which reduce transmission time by 6G technology.

The propagation latency is significantly reduced to 0.05 *ms* in 6G compared to 5G, due to the use of MIMO and Short-range THz. In 6G, AI-native optimization and quantum-inspired edge processing have a significant impact on edge-cloud computing, resulting in edge processing latency of only 0.5 *ms*. Similarly, cloud processing delay is approximately half that of 5G usage, ranging from 0 to 20 *ms*. Transport Network Delay is powered by Optical wireless backhaul in 6G, so the Backhaul Latency is only 5 *ms* [109–111]. To provide an in-depth discussion of the analysis of 6G-enabled edge computing, we have studied more than 100 papers, year-wise, as illustrated by the year-wise analysis in Fig. 23.

We have also analyzed the following technologies listed below to illustrate the practical and real-life applications of 6G-enabled edge-fog-cloud computing. The comparative analysis is also given in Table 6. It presents a comparative analysis of 6G's performance based on

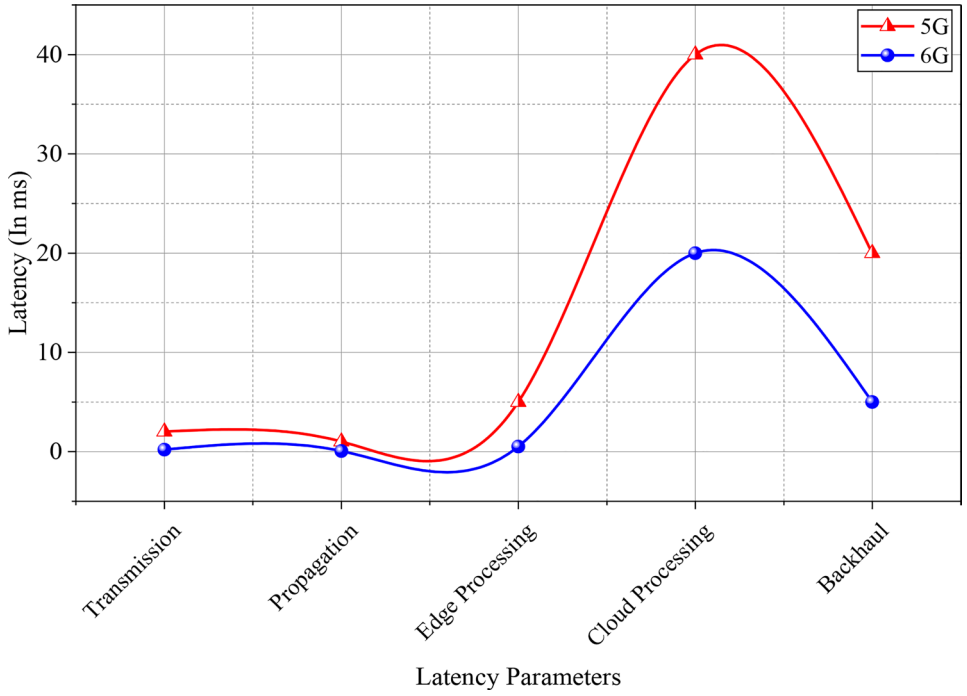


Fig. 22 Futuristic 6G applications

Table 6 Comparative analysis of 6G technology with applications

Authors & Ref.	IoT	Cloud	Block- chain	Agric-ulture	Quantum Computing	Smart City & Industries
Loutfi et al., 2024 [1]	✓	✓	✗	✗	✗	✓
Liu et *al.*, 2024 [11]	✓	✓	✗	✓	✗	✓
Duan et *al.*, 2023 [34]	✓	✓	✓	✗	✗	✓
Nguyen et *al.*, 2024 [74]	✓	✓	✗	✓	✓	✓
Shahraki et *al.*, 2022 [56]	✓	✓	✓	✓	✓	✓
Liao et *al.*, 2021 [45]	✓	✓	✓	✗	✗	✓
Tera et *al.*, 2020 [4]	✓	✓	✗	✗	✓	✓
Nawaz et *al.*, 2019 [37]	✓	✓	✗	✗	✓	✓
Saad et *al.*, 2019 [12]	✓	✓	✓	✓	✓	✓
Baiqing et *al.*, 2019 [5]	✗	✓	✗	✗	✗	✓
Proposed Work	✓	✓	✓	✓	✓	✓



Fig. 24 Future research directions of 6G

the technologies used, as evaluated by various authors. It highlights key factors such as IoT, cloud, blockchain, agriculture, quantum computing, smart cities, and other parameters, indicating which aspects each author addresses.

Future research directions

There persists a number of research gaps in intelligent edge computing, spite of the recent advancements in 6G networks and cloud computing, as illustrated in Fig. 24. These unresolved Challenges call for creative solutions and cooperative investigation, and a number of potential research directions may improve. Some significant future direction based on research gap are as follows:

- Energy Efficient & Carbon Natural Networking Strategies

6G infrastructure's material and energy requirements could greatly raise carbon emissions, so effective and sustainable solutions for real-time job distribution while maintaining dynamic energy trading are needed [112]. This entails maximizing inference workloads, mobility patterns, dynamic energy enabled job distribution, and sensor activation schedules in order to improve system lifetime while preserving mapping accuracy [113].

- Multi Agent Reinforcement Learning

A decentralized edge-cloud architecture rendered feasible by 6G is coupled by multi-agent DRL, which permits effective, distributed resource provisioning across edge nodes [114]. The tightly integrated integration of a reinforcement learning edge-cloud method, cross-layer optimization, and deep learning-based resource allocation are also required. Incorporating ultra-reliable 6G technology for processing real-time applications is another requirement for future research areas in 6G-based edge-cloud computing [115].

- AI Driven End-to-End Orchestration

To cope with the demands of distributed intelligence and real-time adaptation in 6G networks, AI approaches such federated learning, split learning, incremental learning, LLMs, agentic AI, and transfer learning are vital [116]. To increase adaptability, future research ought to concentrate on stable structures such actor-critic ensembles, hybrid learning, and adaptive reward shaping.

- Secure Trustworthy 6G Enabled Edge AI

AI decision-making that lacks transparency undermines compliance and confidence. Robust aggregation,

adversarial training, XAI, zero trust model, post quantum cryptography techniques, and blockchain-based integrity to improve security and reliability are promising research directions [117].

- Scalable FL at the Edge-Cloud Nodes

By detecting malicious activity and guaranteeing robust model performance, anomaly detection is essential to the security of FL in 6G networks [2]. While FL enhances decentralization and anonymity, it has issues with device heterogeneity, and communication cost. Compression-aware techniques, asynchronous updates, and customized FL frameworks are a few potential solutions.

Conclusion

This paper explored cutting-edge technological developments combining edge computing, 6G networks, and cloud computing in many sectors. Edge computing significantly decreased latency and made real-time capabilities possible by processing data closer to the source. These features were essential for applications like industrial automation, smart cities, and driverless cars. This synergy produced an extremely effective and integrated ecosystem when coupled with the scalable storage and processing capacity of cloud computing and the lightning-fast, low-latency connectivity of 6G networks. The paper also highlighted how technologies' convergence promoted innovation in industries like manufacturing, telecommunications, and healthcare by enabling better resource management, smarter decision-making, and more data privacy. More dynamic, adaptable systems were made possible by this integration, which changed digital ecosystems and improved the responsiveness and personalization of technology. These features make cutting-edge technologies highly adaptable for use in finance, healthcare, gaming, and supply chain management, followed by several significant changes that were the main focus of this revolutionary impact's future course. In addition, it also acknowledges the ability of gadgets to make judgments in real-time, improving automation and customization, which was made possible by significant advancements in edge AI and machine learning. With ongoing research focused on creating more reliable, decentralized models, security and privacy continued to be a top goal.

The paper also studied how more remote and underserved areas could utilize edge computing as 6G networks grew, giving everyone equal access to high-speed, low-latency services. Additionally investigated was the integration of quantum computing with edge-cloud models, which holds promise for additional data processing and management optimization. In the end, the combination of edge computing, 6G, and cloud computing transformed industry operations and ushered in a new technological era with enormous potential for innovation and social change.

Author contributions
The idea and problem formulation along with proposed solution, result analysis, and by corresponding author & supervisor, and verifies by all other authors.

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Data availability
The data set generated and/or analyzed during the current study is available upon reasonable request from the corresponding author. However, data sets are available as open source(s).

Declarations

Ethics approval and consent to participate
The work is not submitted in any other journal. There is no conflict of interest. All authors agree to publish.

Competing interests
The authors declare no competing interests.

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References

1. Loutfi SI, Shayea I, Tureli U, A.El-Saleh A, Tashan W (2024) An overview of mobility awareness with mobile edge computing over 6G network: challenges and future research directions. Results Eng 23(102601):1–18
2. Jain M, Verma R, Kumar S, Kumar G, Chamola V, Guizani M (2025) Unveiling the future: a comprehensive analysis of 6G technology and its transformative potential. Telecommunication Syst 88(91):1–38
3. Adhikari M, Hazra A, Menon VG, Chaurasia BK, Mumtaz S (2021) A Roadmap of next-generation wireless technology for 6G-enabled Vehicular networks. IEEE Internet Things Mag 4(4):79–85
4. Tera SP, Chinthaginjala R, Pau G, Kim TH (2024) Toward 6G: an overview of the next generation of intelligent network connectivity. In IEEE Access, vol 13. pp 925–961
5. Fan ZBC, Wang X, Duan X, Wang B, Wang J (2019) 6G technologies: key drivers, core requirements, system architectures, and enabling technologies. IEEE Veh Technol Mag 18(3):1–27
6. Yuan A, Yang Z, Sun Z (2025) Evolution of satellite communication systems Toward 5G/6G for 2030 and beyond. Engineering 1–14
7. Letaief KB, Chen W, Shi Y, Zhang J, Zhang YJA (2019) The Roadmap to 6G: AI empowered wireless networks. IEEE Commun Mag 57(8):84–90
8. Tariq F, Khandaker MRA, Wong KK, Imran M, Bennis M, Debbah M (2020) A speculative study on 6G. IEEE Wireless Commun 27(4):118–125
9. Chamola V, Peelam MS, Guizani M, Niyato D (2025) Future of connectivity: a comprehensive review of innovations and challenges in 7G smart networks. IEEE Open J The Commun Soc 6:3555–3613
10. Qadir Z, Le KN, Saeed N, Munawar HS (2023) Towards 6G Internet of Things: recent advances, use cases, and open challenges. ICT Express 9(3):296–312
11. Liu Y, Deng Y, Nallanathan A, Yuan J (2023) Machine learning for 6G enhanced ultra-reliable and low-latency services. IEEE Wireless Commun 30(2):48–54
12. Saad W, Bennis M, Chen M (2019) A vision of 6G wireless systems: applications, trends, technologies, and open research problems. IEEE Network 34(3):134–142
13. Petkova R, Bozhilov I, Manolova A, Tonchev K, Poulkov V (2024) On the way to holographic-type communications: perspectives and enabling technologies. In IEEE Access, vol 12. pp 59236–59259
14. Porambage P, Gür G, Osorio DPM, Liyanage M, Gurtov A, Ylianttila M (2021) The Roadmap to 6G security and privacy. IEEE Open J The Commun Soc 2:1094–1122

15. Xu T, Wang N, Pang Q, Zhao X (2024) Security and privacy of 6g wireless communication using fog computing and multi-access edge computing. Scalable Comput: Pract Exper 25(2):770–781
16. Wang M, Zhu T, Zhang T, Zhang J, Yu S, Zhou W (2020) Security and privacy in 6G networks: new areas and new challenges. Digit Commun Networks 6(3):281–291
17. Akbar MA, Khan AA, Hyrynsalmi S (2024) Role of quantum computing in shaping the future of 6G technology. "In Inf Softw Technol 170(107454):1–15
18. Serôdio C, Cunha J, Candela G, Rodriguez S, Sousa XR, Branco F (2023) The 6G ecosystem as support for IoE and private networks: vision, requirements, and challenges. Future Internet 15(348):1–32
19. Sanjalawe Y, Fraihat S, Abualhaj M, Makhadmeh S, Alzubi E (2025) A review of 6G and AI convergence: enhancing communication networks with artificial intelligence. IEEE Open J The Commun Soc 6:2308–2355
20. Stergiou CL, Psannis KE, Gupta BB (2020) IoT-based big data secure management in the fog over a 6G wireless network. IEEE Internet Things J 8(7):5164–5171
21. Pournazari J, Ullah A, Al-Dubai A, Liu X (2025) Computation offloading in the edge-to-cloud compute continuum: a survey of federated architectural solutions. Cluster Comput 28(839):1–41
22. Nencioni G, Garroppo RG, Olimid RF (2023) 5G multi-Access edge computing: a survey on security, dependability, and performance. IEEE Access 11:63496–63533
23. Zhao L, Zhou G, Zheng G, I C-L, You X, Hanzo L (2021) Open-source multi-Access edge computing for 6G: opportunities and challenges. IEEE Access 9:158426–158439
24. Jain M, Verma R, Kumar SL, Kumar G, Chamola V, Guizani M (2025) Unveiling the future: a comprehensive analysis of 6G technology and its transformative potential. Telecommunication Syst 88(91):1–38
25. Hakeem SAA, Hussein HH, Kim HW (2022) Vision and research directions of 6G technologies and applications. J King Saud Univ - Comput Inf Sci 34(6):2419–2442
26. Ergena M, Saouda B, Shayeaa I, El-Salehd AA, Ergena O, Inanf F, Tuysuz MF (2024) Edge computing in future wireless networks: a comprehensive evaluation and vision for 6G and beyond. ICT Express 10:1151–1173
27. Pote XR, Hemavathi R, Kumar NA, Santhosh S, Krishnan T, Rajendran VP (2025) Data Protection and security management in the 6G era: addressing high density cloud computing challenges. J Mach Comput 14–27
28. Mallikarjunaradhya V, Yennapusa H, Palle RR, Suganyadevi K, Gupta N (2024) Impacts of high denisty cloud computing on data Protection and security management for 6G Networking. In 2nd International Conference on Disruptive Technologies (ICDT), pp 617–622
29. Tang X, Cao C, Wang Y, Zhang S, Liu Y, Li M, He T (2021) Computing power network: the architecture of Convergence of computing and Networking towards 6G requirement. China Commun 18(2):175–185
30. Shen X, Gao J, Wu W, Lyu K, Li M, Zhuang W, Li X, Rao J (2020) AI-assisted network-slicing based next-generation wireless networks. IEEE Open J Veh Technol 1(1):45–66
31. Wu W, Zhou C, Li M, Wu H, Zhou H, Zhang N, Shen X, Zhuang W (2022) AI-Native network slicing for 6G networks. IEEE Wireless Commun 29(1):96–103
32. Ziegler V, Viswanathan H, Flinck H, Hoffmann M, Räisänen V, Hätönen K (2020) 6G architecture to connect the worlds. In IEEE Access, vol 8. pp 173508–173520
33. Wang S, Zhang X, Zhang Y, Wang L, Yang J, Wang W (2017) A survey on mobile edge networks: convergence of computing, caching and communications. In IEEE Access, vol 5. pp 6757–6779
34. Duan Q, Huang J, Hu S, Deng R, Lu Z, Yu S (2023) Combining federated learning and edge computing toward ubiquitous intelligence in 6G network: challenges, Recent advances, and future directions. IEEE Commun Surv Tutorials 25(4):2892–2950
35. Sefati SS, Haq AU, Craciunescu R, Halunga S, Mihovska A, Fratu O (2024) A comprehensive survey on resource management in 6G network based on Internet of Things. In IEEE Access, vol 12. pp 113741–113784
36. Oukebdane MA, Shah AS, Azad AK, Ekoru J, Madahana M (2025) Unraveling the nexus of ML and 6G: challenges, opportunities, and future directions. In IEEE Access, vol 13. pp 114934–114958
37. Nawaz SJ, Sharma SK, Wyne S, Patwary MN, Asaduzzaman M (2019) Quantum machine learning for 6G communication networks: state-of-the-art and vision for the future. In IEEE Access, vol 7. pp 46317–46350
38. Peelam MS, Rout AA, Chamola V (2024) Quantum computing applications for Internet of Things. IET Quantum Commun 5:103–112
39. Alzaabi FR, Mehmood A (2024) A review of recent advances, challenges, and opportunities in malicious insider threat detection using machine learning methods. In IEEE Access, vol 12. pp 30907–30927
40. Patra A, Pandey A, Hassija V, Chamola V, Mishra RP (2024) A survey on edge enabled metaverse: applications, technological innovations, and prospective trajectories with in the industry. In IEEE Access, vol 12. pp 125125–125144
41. Ali ZH, Ali HA (2022) EEOMA: End-to-end oriented management architecture for 6G-enabled drone communications. Peer-To-Peer Netw Appl 15:1232–1254
42. Han C, Wang Y, Abbasi NA, Li Y, Chen Y, Kürner T, Molisch AF (2022) Terahertz wireless channels: a holistic survey on measurement, modeling, and analysis. IEEE Commun Surv Tutorials 24(3):1670–1707
43. Sadatdiynov K, Cui L, Zhang L, Huang JZ, Salloum S, Mahmud MS (2023) A review of optimization methods for computation offloading in edge computing networks. Digit Commun Networks 9:450–461
44. Essaili AE, Thorson S, Jude A, Ewert JC, Tyudina N, Caltenco H, Litwic L, Burman B (2022) Holographic communication in 5G networks. Holographic Commun 5G Networks 5:2–11
45. Liao S, Wu J, Li J, Konstantin K (2020) Information-centric massive IoT-based ubiquitous connected VR/AR in 6G: a proposed caching consensus approach. IEEE Internet Things J 8(7):5172–5184
46. Sahin E, Stoykova E, Mäkinen J, Gotchev A (2020) Computer-generated holograms for 3D imaging: a survey, "in ACM Comput. Survey 53(32):1–35
47. Sai S, Shah K, Chamola V (2025) A novel diffusion-guided semantic data augmentation method for vehicular IoT. In IEEE Internet of Things Magazine, pp 1–8
48. Cook DJ, Das SK (2007) How smart are our environments? An updated look at the state of the art. Pervasive Mob Comput 3(2):53–73
49. Yadav VK, Andola N, Verma S, Venkatesan S (2023) PSCLS: provably secure certificateless signature scheme for IoT device on cloud. The J Supercomput 79:4962–4982
50. Singh H, Yadav P, Rishiwal V, Yadav M, Tanwar S, Singh O (2025) Localization in WSN-Assisted IoT networks using machine learning techniques for smart agriculture. Int J Commun Syst 38(5):e6004
51. Popovski P, Trillingsgaard KF, Simeone O, Durisi G (2018) 5G wireless network slicing for eMBB URLLC and mMTC: a communication-theoretic view. In IEEE Access, vol 6. pp 55765–55779
52. Nwakanma CI, Anantha AP, Islam FB, Lee JM, Kim DS (2020) 3GPP release-16 for industrial Internet of things and mission critical communications. In International Conference on Information and Communication Technology Convergence (ICTC), pp 403–406
53. Nwadiugwu WP, Nwakanma CI, Lee JM, Kim DS (2019) Achieving reliable URLLC-based network in industrial and military embedded systems. In International Conference on Information and Communication Technology Convergence (ICTC), pp 642–645
54. Anand A, Veciana GD, Shakkottai S (2020) Joint scheduling of URLLC and eMBB traffic in 5G wireless networks. "In IEEE/ACM Trans Netw 28(2):477–490
55. Yang X, Zho Z, Huang B (2021) URLLC key technologies and standardization for 6G power Internet of Things. IEEE Commun Stand Mag 5(2):52–59
56. Shahraki A, Abbasi M, Piran MJ, Taherkordi A A comprehensive survey on 6G networks: applications core services enabling technologies and future challenges, Online available at: https://arxiv.org/pdf/2206.00868. Last Accessed:11 Sept., 2025
57. Alsharif MH, Jahid A, Kannadasan R, Kim M-K (2024) Unleashing the potential of sixth generation (6G) wireless networks in smart energy grid management: a comprehensive review. Energy Rep 11:1376–1398
58. Shi Y, Yang K, Jiang T, Zhang J, Letaief KB (2020) Communication-efficient edge AI: algorithms and systems. IEEE Commun Surv Tutorials 22(4):2167–2191
59. Zhou Z, Chen X, Li E, Zeng L, Luo K, Zhang J (2019) Edge intelligence: paving the last mile of artificial intelligence with edge computing. Proc The IEEE 107(8):1738–1762
60. Tao F, Zhang H, Liu A, Nee AYC (2019) Digital twin in industry: state-of-the-art. IEEE Trans Ind Inf 15(4):2405–2415
61. Qi Q, Tao F, Hu T, Anwer N, Liu A, Wei Y, Wang L, Nee AYC (2019) Enabling technologies and tools for digital twin. J Manuf Syst 58(Part B):3–21
62. Lin Y, Chen L, Ali A, Nugent C, Cleland I, Li R, Ding J, Ning H (2024) Human digital twin: a survey. J Cloud Comput Adv Syst Appl 13(131):1–21
63. Soliman MM, Ahmed E, Darwish A, Hassanien AE (2024) Artificial intelligence powered metaverse: analysis, challenges and future perspectives. Artif Intel Rev 57(36):1–46

64. Kadam PA, Bhushan S, Rishiwal V, Yadav M, Yadav P, Agarwal U (2025) ValidCertify: an effective blockchain-based domain certificate authentication and verification scheme. Peer-To-Peer Netw Appl 18(177):1–18
65. Ghosh A, Lavanya VH, Chamola V, Saddik AE (2024) A survey on decentralized metaverse using Blockchain and Web 3.0 technologies, applications, and more. In IEEE Access, pp 1–34
66. Pajooh HH, Demidenko S, Aslam S, Harris M (2022) Blockchain and 6G-Enabled IoT. Inventions 7(109):1–18
67. Varadarajan MN, Seeni SK (2024) Innovative digital ownership and collectibles via proof of stake (POS) and non-fungible tokens (NFTS). Int J Adv Signal Image Sci 10(1):22–34
68. Hewa T, Gür G, Kalla A, Ylianttila M, Bracken A, Liyanage M (2022) The role of Blockchain in 6G: challenges, opportunities and research directions. In Proceedings of the 2020 2nd 6G Wireless Summit (6G SUMMIT), pp 1–5
69. Sengupta J, Ruj S, Bit SD (2020) A comprehensive survey on attacks, security issues and Blockchain solutions for IoT and IIoT. J Network Comput Appl 149(102481):1–20
70. Gupta S, K P, Srivastava NV, Kumar A, Chaurasia BK (2024) Heart disease prediction using federated learning, In: Singh Y, Gonçalves PJS, Singh PK, Kolekar MH (eds) Proceedings of International Conference on Recent Innovations in Computing. ICRIC 2023. Lecture Notes in Electrical Engineering, vol 1196. Springer, Singapore, 35–47
71. Blika A, Palmos S, Doukas G, Lamprou V, Pelekis S, Kontoulis M, Ntanos C, Askounis D (2024) Federated learning for enhanced cybersecurity and trustworthiness in 5G and 6G networks: a comprehensive survey. In IEEE Open J Commun Soc, pp 1–42
72. Li T, Sahu AK, Talwalkar A, Smith V (2020) Federated learning: challenges, methods, and future directions. IEEE Signal Process Mag 37(3):50–60
73. Subedi P, Alsadoon A, Prasad PWC, Rehman S, Giweli N, Imran M, Arif S (2021) Network slicing: a next generation 5G perspective. EURASIP J Wireless Commun Netw 102:1–26
74. Nguyen HT, Krishnan P, Krishnaswamy D, Usman M, Buyya R (2024) Quantum cloud computing: a review, open problems, and future directions 1–35, Online available at: arXiv preprint arXiv: 2404.11420, Last accessed: 11 Sept. 2025
75. Duong TQ, Nguyen LD, Narottama B, Ansere JA, Huynh DV, Shin H (2022) Quantum-inspired real-time optimisation for 6G networks: opportunities challenges and the road ahead. IEEE Open J. Commun. Soc 3:1347–1359
76. Singh P, Dasgupta R, Singh A, Pandey H, Hassija V, Chamola V, Sikdar B (2024) A survey on available tools and technologies enabling quantum computing. In IEEE Access, vol 12. pp 57974–57991
77. Alex SA, Menon VG, Adhikari M, Mumtaz S (2025) Zero-touch self-organizing networks in 6G: integrating edge computing and Artificial intelligence. IEEE communications standards magazine, 2471–2825
78. Luong NC, Jiao Y, Wang P, Niyato D, Kim DI, Han Z (2020) A machine-learning-based auction for resource trading in fog computing. IEEE Commun Mag 58(3):82–88
79. Zong B, Fan C, Wang X, Duan X, Wang B, Wang J (2019) 6G technologies: key Drivers, core requirements, system architectures, and enabling technologies. IEEE Veh Technol Mag 14(3):18–27
80. Jawad AT, Maaloul R, Chaari L (2023) A comprehensive survey on 6G and beyond: enabling technologies, opportunities of machine learning and challenges. Comput Networks 237:110085
81. Alotaibi A, Barnawi A (2023) Securing massive IoT in 6G: recent solutions, architectures, future directions. Internet Things 22:100715
82. Kharel B, López PLA, Mahmood NH, Alves H, Latva-Aho M (2021) Fog-RAN enabled multi-connectivity and multi-cell scheduling framework for ultra-reliable low latency communication. In IEEE Access, vol 10. pp 7059–7072
83. Checko A, Christiansen HL, Yan Y, Scolari L, Kardaras G, Berger MS, Dittmann L (2014) Cloud RAN for mobile networks-a technology overview. IEEE Commun Surv Tutorials 17(1):405–426
84. Javed AR, Shahzad F, Rehman SR, Zikria YB, Razzak I, Jalil Z, Xu G (2022) Future smart cities: requirements, emerging technologies, applications, challenges, and future aspects. Cities 129(103794):1–50
85. Yang H, Alphones A, Xiong Z, Niyato D, Zhao J, Wu K (2020) Artificial intelligence-enabled intelligent 6G networks. IEEE Network 34(6):272–280
86. Alsamhi SH, Shvetsov AV, Kumar S, Hassan J, Alhartomi MA, Shvetsova SV, Sahal R, Hawbani A (2022) Computing in the sky: a survey on intelligent ubiquitous computing for UAV-Assisted 6G networks and industry 4.0/5.0. Drones 6(177):1–29
87. Gebresenbet G, Bosona T, Patterson D, Persson H, Fischer B, Mandaluniz N, Chirici G, Zacepins A, Komasilovs V, Pitulac T, Nasirahmadi A (2023) A concept for application of integrated digital technologies to enhance future smart agricultural systems. Smart Agric Technol 5(100255):1–12
88. Tomaszewski L, Kołakowski R (2023) Mobile services for smart agriculture and forestry, biodiversity monitoring, and water management: challenges for 5G/6G networks. Telecom 4:67–99
89. Nova SN, Rahman MS, Chakraborty C (2021) Patient's health surveillance model using IoT and 6G technology. Green Technol Innov Sustain Smart Soc 191–209
90. Rauschnabel PA, Felix R, Hinsch C, Shahab H, Alt F (2022) What is XR? Towards a framework for augmented and virtual reality. Comput Hum Behav 133(107289):1–18
91. Ghosh A, Hassija V, Chamola V, Saddik AE (2024) A survey on decentralized metaverse using Blockchain and Web 3.0 technologies, applications, and more. In IEEE Access, vol 12. pp 146915–146948
92. Zakarneh B, Annamalai N, Alquqa EK, Mohamed KM, Salhi NRA (2024) Virtual reality and alternate realities in Neal Stephenson's ——Snow Crash. World J Engl Language 14(2):244–252
93. Ball M (2022) The metaverse: and how it will revolutionize everything. Liveright Publishing 292. Online available at: https://www.tandfonline.com/doi/full/10.1080/15228053.2022.2136927. Last Accessed 23 Sept. 2025
94. Yu Q, Ren J, Zhou H, Zhang W (2020) A cybertwin based network architecture for 6G. In IEEE 2nd 6G Wireless Summit (6G SUMMIT), pp 1–5
95. Lucas-Estañ MC, Gozalvez J (2022) Sensing-based grant-free scheduling for ultra reliable low latency and deterministic beyond 5G networks. IEEE Trans Veh Technol 71(4):4171–4183
96. Jiang F, Wang K, Dong L, Pan C, Xu W, Yang K (2019) Deep-learning-based joint resource scheduling algorithms for hybrid MEC networks. IEEE Internet Things J 7(7):6252–6265
97. Shafie A, Yang N, Han C, Jornet JM, Juntti M, Kürner T (2022) Terahertz communications for 6g and beyond wireless networks: challenges, key advancements, and opportunities. IEEE Network 37(3):162–169
98. M.Othman W, Ateya AA, Nasr ME, Muthanna A, Affendi ME, Koucheryavy A, Hamdi AA (2025) Key enabling technologies for 6G: the role of UAVs, Terahertz communication, and intelligent Reconfigurable Surfaces in shaping the future of wireless networks. J Sens Actuator Networks 14(30):1–74
99. Atre A, Singh KK, Chaurasia BK, Sengar SS, Srivastava P (2025) Optimized resource analysis in VANET cloud using fog enabled real-time traffic control System. Proc Eng Sci 7(3):1939–1954
100. Shi W, Cao J, Zhang Q, Li Y, Xu L (2016) Edge computing: vision and challenges. IEEE Internet Things J 3(5):637–646
101. Basar E, Alexandropoulos GC, Liu Y, Wu Q, Jin S, Yuen C, Dobre OA, Schober R (2024) Reconfigurable intelligent Surfaces for 6G: emerging hardware Architectures, applications, and open challenges. IEEE Veh Technol Mag 19(3):27–47
102. Zhao X, Jian M, Chen Y, Zhao Y, Mu L (2025) Reconfigurable intelligent surfaces for 6G: engineering challenges and the road ahead. Intell Converged Networks 6(1):53–81
103. Letaief KB, Shi Y, Lu J, Lu J (2022) Edge Artificial intelligence for 6G: vision, enabling technologies, and applications. IEEE J Sel Areas Commun 40(1):5–36
104. Alimi IA et al (2021) 6G CloudNet: towards a distributed, autonomous, and Federated AI-enabled cloud and edge computing. In: Wu, Y., true etal 6G Mobile wireless networks. Computer communications and networks. Springer, Cham, pp 251–283
105. He Q, Lin J, Fang H, Wang X, Huang M, Yi X, Yu K (2025) Integrating IoT and 6G: applications of edge intelligence, challenges, and future directions. IEEE Trans Serv Comput 18(4):2471–2488
106. Heydarishahreza N, Han T, Ansari N (2025) Spectrum sharing and interference management for 6G LEO satellite-terrestrial network integration. IEEE Commun Surv Tutorials, Published In: IEEE Commun Surv Tutorials 27(5):2794–2825
107. Jayachandran M, Gatla RK, Flah A, Milyani AH, Milyani HM, Blazek V, Prokop L, Kraiem H (2024) Challenges and opportunities in green hydrogen adoption for decarbonizing hard-to-abate industries: a comprehensive review. IEEE Access 12:23363–23388
108. Sharma AK, Peelam MS, Chaurasia BK, Chamola V (2024) QIoTChain: quantum IoT-Blockchain fusion for advanced data Protection in industry 4.0. IET Blockchain 4(3):252–262
109. Popovski P, Chiariotti F, Huang K, Kalør AE, Kountouris M, Pappas N (2022) A perspective on time toward wireless 6G. Proc The IEEE 110(8):1116–1146
110. Mohammed SM, Al-Barrak A, Mahmood NT (2024) Enabling technologies for ultra-low latency and high-reliability communication in 6G networks. Inf Syst Eng 29(3):1195–1208

111. Hazra A, Munusamy A, Adhikari M, Awasthi LK (2024) 6G-Enabled ultra-reliable low latency communication for industry 5.0: challenges and future directions. IEEE Commun Stand Mag 8(2):36–42
112. Zhang P, Xiao Y, Li Y, Ge X, Shi G, Yang Y (2023) Towards net-zero carbon emissions in network AI for 6G and Beyond. IEEE Commun Mag 62(4):58–64
113. Wang X, Tang Z, Guo J, Meng T, Wang C, Wang T, Jia W (2025) Empowering edge intelligence: a comprehensive survey on on-device ai models. ACM Comput Surv 57(9):1–39
114. Tran D-H, Waheed N, Saputra YM, Lin X, Nguyen CT, Abdu TS, Vo VN, Pham V-Q, Alsenwi M, Adam ABM, Chatzinotas S, Lagunas E, Tran H, Dac TH, Huynh NV (2025) Network digital twin for 6G and Beyond: an end-to-end view across multi-domain network ecosystems. IEEE Open J The Commun Soc 6:6866–6911
115. Adhikari M, Hazra A (2022) 6G-Enabled ultra-reliable low-latency communication in edge networks. EEE Commun Stand Mag 6(1):67–74
116. Yang H, Liu H, Yuan X, Wu K, Ni W, Zhang JA, Liu RP (2025) Synergizing intelligence and privacy: a review of integrating Internet of Things, large Language models, and Federated learning in advanced networked systems. Appl Sci 15(6587):1–34
117. Peelam MS, Chamola V, Sai S, Jalan P (2025) Blockchain-based game theoretical framework for V2V and V2G energy trading in carbon-intelligent Internet of vehicles. IEEE Internet Things J 12(19):40034–40053

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